

The rise and potential of large language model based agents: a survey

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Abstract For a long time, researchers have sought artificial intelligence (AI) that matches or exceeds human intelligence. AI agents, which are artificial entities capable of sensing the environment, making decisions, and taking actions, are seen as a means to achieve this goal. Extensive efforts have been made to develop AI agents, with a primary focus on refining algorithms or training strategies to enhance specific skills or particular task performance. The field, however, lacks a sufficiently general and powerful model to serve as a foundation for building general agents adaptable to diverse scenarios. With their versatile capabilities, large language models (LLMs) pave a promising path for the development of general AI agents, and substantial progress has been made in the realm of LLM-based agents. In this article, we conduct a comprehensive survey on LLM-based agents, covering their construction frameworks, application scenarios, and the exploration of societies built upon LLM-based agents. We also conclude some potential future directions and open problems in this flourishing field.

Keywords natural language processing, large language models, LLM-based agents, AI agents, agent society

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1 Introduction

One core research area in artificial intelligence (AI) is to develop agents that possess human-level intelligence, and potentially even exceed it [1]. The concept of agent originates in philosophy, where it describes entities possessing desires, beliefs, intentions, and the ability to take actions [2]. In AI research, the term agent refers to an artificial entity capable of perceiving its surroundings using sensors, making decisions, and then taking actions in response using actuators [1, 3]. The development and advancement of agents have been central within the AI community for a long time [1, 4]. Moreover, AI agents are now recognized as a pivotal stride towards achieving artificial general intelligence (AGI)¹⁾, as they encompass the potential for a wide range of intelligent activities [3, 5, 6].

From the mid-20th century, significant strides were made in designing and developing AI agents [7–12]. However, these efforts have predominantly focused on enhancing specific skills (such as symbolic reasoning or quick responsiveness) or mastering particular tasks (such as Go or Chess) [13–15]. Achieving broad adaptability across varied scenarios remained elusive. Moreover, previous studies mostly emphasized the design of algorithms and training strategies, overlooking the development of the model's inherent general

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1) Also known as Strong AI.

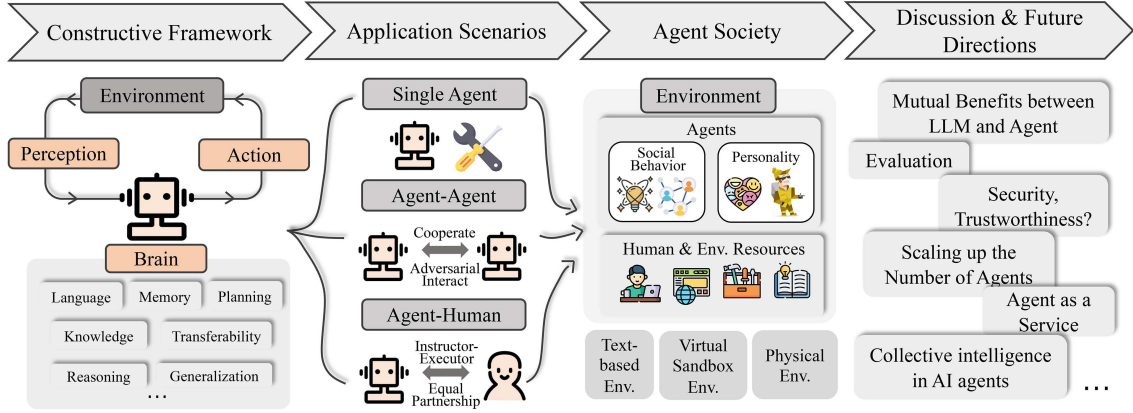


Figure 1 (Color online) Roadmap of this survey.

abilities [16,17]. Actually, enhancing these inherent abilities of the model is pivotal for advancing the agent, and the field is in need of a powerful and versatile model to serve as the foundation for agent systems.

The emergence of large language models (LLMs) has brought a glimmer of hope for the further development of agents [18–20], and significant progress has been made by researchers in the field [16,21–23]. This is attributed to the remarkable capabilities demonstrated by LLMs, including natural language interaction, knowledge acquisition, instruction following, generalization, reasoning, planning, and tool-using. These advantages have earned LLMs the designation of sparks for AGI [24], making them highly desirable for building general agents [16]. And the area of LLM-based agents has emerged as one of the most promising fields in AI research. The academic and industrial communities have invested significant research and development efforts in this field. However, despite the progress, this emerging field still faces numerous crucial issues that require further exploration and resolution, e.g., the architectural design of LLM-based agents and their application domains. Therefore, it is necessary to review the origins and development of this field, summarize current research achievements, and look ahead to provide a clearer understanding of LLM-based agents and deeper insights into their future development.

In this article, we present a comprehensive and systematic survey of LLM-based agents, attempting to explore several critical research problems within the field and prospective avenues in this burgeoning field (Figure 1 demonstrates the roadmap of this survey). Before delving into the research problems, we first discuss the background information (Section 2), including the origin, definition, and development of AI agents, as well as why LLMs are suitable as the foundation for AI agents. The first research problem is the design and construction of LLM-based agents (Section 3). Based on the definition of AI agents, we propose a conceptual framework with three key components: brain, perception, and action. Different downstream applications can tailor this framework according to their specific requirements. The second research problem is the application scenarios of LLM-based agents (Section 4). We discuss the various paradigms such as single agent, multi-agent, and human-in-the-loop, and their respective applicable scenarios. The third research problem is agent society (Section 5). We discuss the components and operational mechanisms of an agent society, as well as its insights into human society. Finally, we discuss a series of open problems and future directions in this field (Section 6), e.g., the mutual benefits between LLM research and agent research, evaluation for LLM-based agents, potential risks of LLM-based agents, and scaling up the number of agents. Finally, we present an overview and summary of the entire article (Section 7).

2 Background

In this section, we provide crucial background information to lay the groundwork for the subsequent content. We first discuss the origin of AI agents in philosophy and its definition (Subsection 2.1). Subsequently, we review the development of AI agents through the lens of technological trends (Subsection 2.2). Finally, we introduce the key characteristics of agents and demonstrate why LLMs are suitable to serve as the foundation of AI agents (Subsection 2.3).

2.1 Origin of AI agent

“Agent” is a concept with a long history that spans across various fields. It is derived from a philosophical origin, triggering a discussion on whether artificial products can possess agency in a philosophical sense. Then, related concepts were introduced into the field of AI, forming the basis of AI agents.

Agent in philosophy. The initial idea of an “agent” has a philosophical origin, with influential thinkers like Aristotle and Hume, contributing to its conceptualization [2]. In a general sense, an “agent” is an entity with the capacity to act, and the term “agency” denotes the exercise or manifestation of this capacity [2]. In a narrow sense, “agency” is usually used to refer to the performance of intentional actions; and correspondingly, “agent” denotes entities that possess desires, beliefs, intentions, and the ability to act [25–28]. Importantly, agents encompass not only individual human beings but also other entities in both the physical and virtual world.

Introduction of agents into AI. In AI, an agent is defined as an artificial entity capable of perceiving its environment, making decisions, and taking actions using sensors and actuators [1, 3]. As Wooldridge et al. [3] stated that we can define AI by saying that it is a subfield of computer science that aims to design and build computer-based agents that exhibit aspects of intelligent behavior. So we can treat “agent” as a central concept in AI. When the concept of agent is introduced into AI, its meaning evolves. In philosophy, an agent can be a human, an animal, or even a concept or entity with autonomy [2]. However, in the field of AI, an agent is a computational entity [3, 29]. Since it is difficult to determine if they possess internal desires or consciousness, many AI researchers, including Alan Turing, suggest temporarily setting aside the question of whether an agent is “actually” thinking or literally possesses a “mind” [30]. Instead, researchers employ other attributes to help describe an agent, such as properties of autonomy, reactivity, pro-activeness, and social ability [3, 31].

It might come as a surprise that researchers within the mainstream AI community devoted relatively minimal attention to concepts related to agents until the mid to late 1980s. Nevertheless, there has been a significant surge of interest in this topic within the realms of computer science and artificial intelligence communities since then [32–35].

2.2 Technological trends in agent research

The evolution of AI agents has undergone several main stages, and here we take the lens of technological trends to review its development briefly.

Symbolic agents. In the early stages of AI research, the predominant approach utilized is symbolic AI, characterized by its reliance on symbolic logic [36, 37]. This approach employs logical rules and symbolic representations to encapsulate knowledge and facilitate reasoning processes. Early AI agents are built based on this approach [38], and they primarily focused on two problems: the transduction problem and the representation/reasoning problem [39]. These agents aim to emulate human thinking patterns. They possess explicit and interpretable reasoning frameworks, and due to their symbolic nature, they exhibit a high degree of expressive capability [7, 8, 40]. A classic example of this approach is knowledge-based expert systems. However, symbolic agents face limitations in handling uncertainty and large-scale real-world problems [13, 14]. Additionally, due to the intricacies of symbolic reasoning algorithms, it is challenging to find an efficient algorithm capable of producing meaningful results within a finite timeframe [14, 41].

Reactive agents. Different from symbolic agents, reactive agents do not use complex symbolic reasoning. Instead, they primarily focus on the interaction between the agent and its environment, emphasizing quick and real-time responses [9, 10, 14, 42, 43]. These agents are mainly based on a sense-act loop, efficiently perceiving and reacting to the environment. The design of such agents prioritizes direct input-output mappings rather than intricate reasoning and symbolic operations [34]. However, reactive agents also have limitations. They typically require fewer computational resources, enabling quicker responses, but they might lack complex higher-level decision-making and planning capabilities.

Reinforcement learning-based agents. With the improvement of computational capabilities and data availability, along with a growing interest in simulating interactions between intelligent agents and their environments, researchers have begun to utilize reinforcement learning methods to train agents for tackling more challenging and complex tasks [11, 12, 44, 45]. The primary concern in this field is how to enable agents to learn through interactions with their environments, enabling them to achieve maximum cumulative rewards in specific tasks [15]. Initially, reinforcement learning (RL) agents are primarily based on fundamental techniques such as policy search and value function optimization, exemplified by

Q-learning [46] and SARSA [47]. With the rise of deep learning, the integration of deep neural networks and reinforcement learning, known as deep reinforcement learning (DRL), has emerged [48, 49]. This allows agents to learn intricate policies from high-dimensional inputs, leading to numerous significant accomplishments like AlphaGo [50] and DQN [51]. The advantage of this approach lies in its capacity to enable agents to autonomously learn in unknown environments, without explicit human intervention. This allows for its wide application in an array of domains, from gaming to robot control and beyond. Nonetheless, reinforcement learning faces challenges including long training times, low sample efficiency, and stability concerns, particularly when applied in complex real-world environments [15].

Agents with transfer learning and meta learning. Traditionally, training a reinforcement learning agent requires huge sample sizes and long training time, and lacks generalization capability [52–56]. Consequently, researchers introduce transfer learning to expedite an agent’s learning on new tasks [57–59]. Transfer learning reduces the burden of training on new tasks and facilitates the sharing and migration of knowledge across different tasks, thereby enhancing learning efficiency, performance, and generalization capabilities. Furthermore, meta learning has also been introduced to develop agents [60–64]. Meta learning focuses on learning how to learn, enabling an agent to swiftly infer optimal policies for new tasks from a small number of samples [65]. Such an agent, when confronted with a new task, can rapidly adjust its learning approach by leveraging acquired general knowledge and policies, consequently reducing the reliance on a large volume of samples. However, when there exist significant disparities between source and target tasks, the effectiveness of transfer learning might fall short of expectations and negative transfer might occur [66, 67]. Additionally, the substantial amount of pre-training and large sample sizes required by meta learning make it hard to establish a universal learning policy [61, 68].

Large language model-based agents. As LLMs have demonstrated impressive emergent capabilities and have gained immense popularity [18–20, 69], researchers have started to leverage these models to construct AI agents, known as LLM-based agents [16, 21, 22, 70]. Specifically, they employ LLMs as the core component of the brain or controller of these agents and expand their perceptual and action space through strategies such as multimodal perception and tool utilization [71–75]. These LLM-based agents can exhibit reasoning and planning abilities comparable to symbolic agents through techniques like chain-of-thought (CoT) and problem decomposition [76–82]. They can also acquire interactive capabilities with the environment, akin to reactive agents, by learning from feedback and performing new actions [83–85]. Similarly, LLMs undergo pre-training on large-scale corpora and demonstrate the capacity for few-shot and zero-shot generalization, allowing for seamless transfer between tasks without the need to update parameters [69, 86–88]. LLM-based agents have been applied to various real-world scenarios, such as software development [89, 90] and scientific research [91]. Due to their natural language comprehension and generation capabilities, they can interact with each other seamlessly, giving rise to collaboration and competition among multiple agents [89, 90, 92, 93]. Furthermore, some studies suggested that allowing multiple agents to coexist can lead to the emergence of social phenomena [16].

2.3 Why is LLM suitable as the foundation of agent?

As mentioned before, researchers have introduced several properties to help describe and define agents in the field of AI. In this section, we will discuss the key properties, elucidate their relevance to LLMs, and thereby expound on why LLMs are highly suited to serve as the foundation of AI agents.

Autonomy. Autonomy means that an agent operates without direct intervention from humans or others and possesses a degree of control over its actions and internal states [3, 94]. This implies that an agent should not only possess the capability to follow explicit human instructions for task completion but also exhibit the capacity to initiate and execute actions independently. LLMs can demonstrate a form of autonomy through their ability to generate human-like text, engage in conversations, and perform various tasks without detailed step-by-step instructions [95, 96]. Moreover, they can dynamically adjust their outputs based on environmental input, reflecting a degree of adaptive autonomy [17, 21, 85]. Furthermore, they can showcase autonomy through exhibiting creativity like coming up with novel ideas, stories, or solutions that have not been explicitly programmed into them [97, 98]. This implies a certain level of self-directed exploration and decision-making. Applications like Auto-GPT [95] exemplify the significant potential of LLMs in constructing autonomous agents. Simply by providing them with a task and a set of available tools, they can autonomously formulate plans and execute them to achieve the ultimate goal.

Reactivity. Reactivity in an agent refers to its ability to respond rapidly to immediate changes and stimuli in its environment [31]. This implies that the agent can perceive alterations in its surroundings and promptly take appropriate actions. Traditionally, the perceptual space of language models has been confined to textual inputs, while the action space has been limited to textual outputs. Recently, advances in multimodal fusion techniques expand their perceptual space to rapidly process visual and auditory information from the environments [19, 99, 100]. Similarly, it is also feasible to expand the action space of LLMs through embodiment techniques [101, 102] and tool usage [73, 75]. These advancements enable LLMs to effectively interact with the real-world physical environment and carry out tasks within it. One major challenge is that LLM-based agents, when performing non-textual actions, require an intermediate step of generating thoughts or formulating tool usage in the textual form before eventually translating them into concrete actions. This intermediary process consumes time and reduces the response speed. However, this is analogous to human behavioral patterns, where the principle of “think before you act” is observed [103, 104].

Pro-activeness. Pro-activeness denotes that agents do not merely react to their environments; they possess the capacity to display goal-oriented actions by proactively taking the initiative [31]. This property emphasizes that agents can reason, make plans, and take proactive measures in their actions to achieve specific goals or adapt to environmental changes. Although intuitively the paradigm of next token prediction in LLMs may not possess intention or desire, research has shown that they can implicitly generate representations of these states and guide the model’s inference process [105–107]. LLMs have demonstrated a strong capacity for generalized reasoning and planning. By prompting LLMs with instructions like “let’s think step by step”, we can elicit their reasoning abilities, such as logical and mathematical reasoning [76–78]. Similarly, LLMs have shown the emergent ability of planning in forms of goal reformulation [80, 108], task decomposition [79, 109], and adjusting plans in response to environmental changes [81, 110].

Social ability. Social ability refers to an agent’s capacity to interact with other agents, including humans, using agent-communication languages [111]. Large language models exhibit strong natural language interaction abilities like comprehension and generation [17, 112, 113]. Compared to structured languages or other communication protocols, such capability enables them to interact with other models or humans in an interpretable manner, laying the foundation for the social ability for LLM-based agents [16, 89]. Many researchers have demonstrated that LLM-based agents can enhance task performance through social behaviors such as collaboration and competition [89, 92, 114, 115]. By inputting specific prompts, LLMs can also play different roles, thereby simulating the social division of labor in the real world [90]. Furthermore, when we place multiple agents with distinct identities into a society, emergent social phenomena can be observed [16].

3 Birth of an agent: construction of LLM-based agents

The principle of “Survival of the Fittest” [116] highlights the need for cognitive abilities to adapt to the external environment and respond to changes for individual survival. Inspired by this and the definition of AI agents, we present a general conceptual framework of an LLM-based agent composed of three key components: brain, perception, and action (see Figure 2). We first describe the structure and working mechanism of the “brain”, which is the cognitive core of an AI agent (Subsection 3.1). It not only stores knowledge and memories but also undertakes indispensable functions like decision-making, exhibiting the intelligence of an agent. Next, we introduce the perception module (Subsection 3.2). Its core purpose is to broaden the agent’s perception space from a text-only domain to a multimodal sphere, which equips the agent to grasp and utilize information from its surroundings more effectively. Finally, we present the action module designed to expand the action space of an agent (Subsection 3.3). Specifically, the agent is empowered to adapt to environmental changes, provide feedbacks, and even influence and mold the environment.

The framework can be tailored for different application scenarios; i.e., not every specific component will be used in all studies. In general, agents operate in the following workflow: First, the perception module, corresponding to human sensory systems, perceives changes in the external environment and then converts multimodal information into an understandable representation for the agent. Subsequently, the brain module, serving as the control center, engages in activities such as thinking, decision-making, and operations with storage including memory and knowledge. Finally, the action module, corresponding

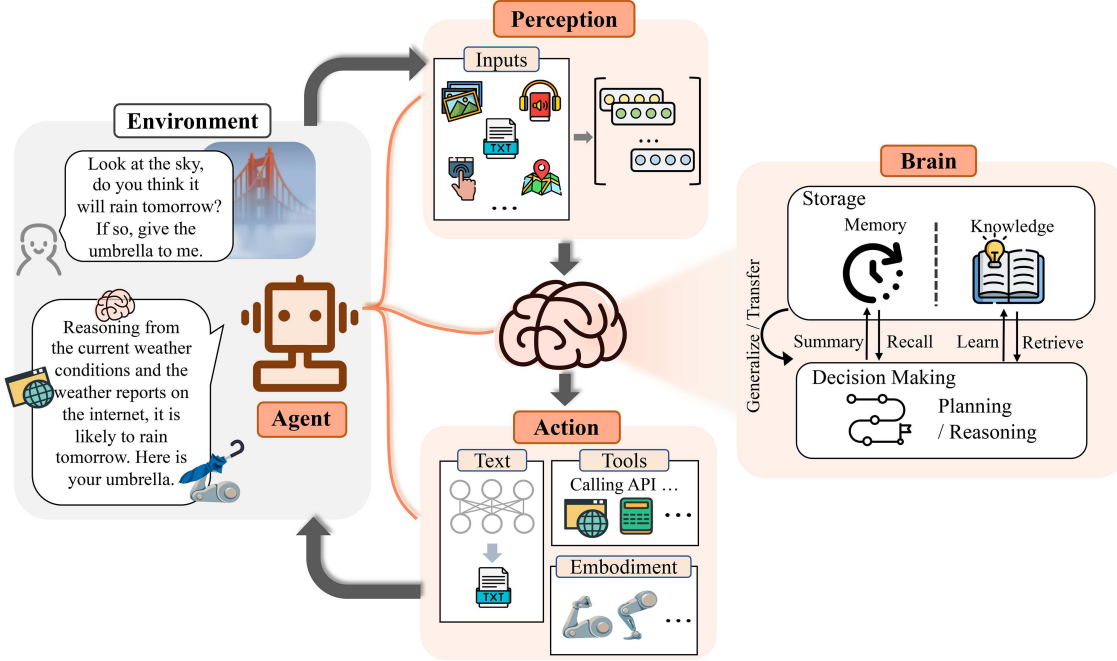


Figure 2 (Color online) Conceptual framework of an LLM-based agent with three components: brain, perception, and action. Serving as the controller, the brain module undertakes basic tasks like memorizing, reasoning, and planning. The perception module perceives and processes multimodal information from the external environment, and the action module carries out the execution and interacts with the surroundings. Here is an example that illustrates the workflow: When a human asks whether it will rain, the perception module converts the instruction into a specific representation. Then the brain module begins to reason according to the current weather and the weather reports on the internet. Finally, the action module responds and hands the umbrella to the human. By repeating the above process, the LLM-based agent can continuously get feedback and interact with the environment.

to human limbs, carries out the execution and leaves an impact on the surroundings. By repeating the above process, an agent can continuously get feedback and interact with the environment.

3.1 Brain

The human brain is a sophisticated structure comprised of a vast number of interconnected neurons, capable of processing complex information, generating diverse thoughts, controlling behaviors, and even creating art and culture [117]. Much like humans, the brain component serves as the central controller of an AI agent. In our framework, the brain module is primarily composed of an LLM.

After receiving the information processed by the perception module, the brain module first turns to storage, retrieving in knowledge (Subsection 3.1.1) and recalling from memory (Subsection 3.1.2). These outcomes aid the agent in planing, reasoning, and making decisions (Subsection 3.1.3). Additionally, the brain module may memorize the agent's past observations, thoughts, and actions in the form of summaries, vectors, or other data structures. Meanwhile, it can also update the knowledge such as common sense and domain knowledge for future use. The LLM-based agent may also adapt to unfamiliar scenarios with its inherent generalization and transferability (Subsection 3.1.4). In the subsequent sections, we delve into a detailed exploration of these extraordinary facets of the brain module.

3.1.1 Knowledge

LLMs are typically imbued with large-scale data through semi-supervised learning during the pre-training phase [118,119]. They can encode a wide range of knowledge into their parameters and respond correctly to various types of queries after pre-training [120]. Furthermore, the knowledge can assist LLM-based agents in making informed decisions [121]. Such knowledge can be broadly categorized into the following types.

- **Commonsense knowledge.** Commonsense knowledge [122–124] refers to general world facts that are typically taught to most individuals at an early age. For example, people commonly know that medicine is used for curing diseases, and umbrellas are used to protect against rain. Such information is

usually not explicitly mentioned in the context. Therefore, models lacking the corresponding common-sense knowledge may fail to grasp or misinterpret the intended meaning [125]. Similarly, agents without commonsense knowledge may make incorrect decisions, such as not bringing an umbrella when it rains heavily.

- **Professional domain knowledge.** Professional domain knowledge refers to the knowledge associated with a specific domain like programming [124, 126, 127], mathematics [128], medicine [129], etc. It is essential for models to effectively solve problems within a particular domain [130]. For example, models designed to perform programming tasks need to possess programming knowledge. This kind of knowledge is very important for domain-specific agents because only with this knowledge can they make reasonable decisions.

3.1.2 *Memory*

In our framework, “memory” stores the agent’s past observations, thoughts, and actions [131]. This memory enables agents to use past experiences for strategy formulation and decision-making, similar to how humans do [132–134]. It assists agents in handling complex problems by allowing them to revisit past strategies and adapt to new environments.

However, as interactions in LLM-based agents increase, two main issues emerge. First, the length of historical data can exceed the processing capacity of Transformer architecture-based LLM agents, leading to possible data truncation. Second, with the accumulation of extensive data, it becomes increasingly difficult for agents to retrieve and connect relevant memories, risking misaligned responses in ongoing contexts. Therefore, appropriate memory compression and retrieval techniques are key to enhancing an agent’s memory capabilities.

Methods for memory compression. We delve into two primary methods designed to compress memory, ensuring efficient recall and analysis.

- **Summarizing memory.** The first strategy for improving memory efficiency is memory summarization. This method summarizes historical interactions and stores them in natural language. Various techniques have been proposed for memory summarization. Using prompts, some methods succinctly integrate memories [135], while others emphasize reflective processes to create condensed memory representations [16, 136]. Hierarchical methods streamline dialogues into both daily snapshots and overarching summaries [137]. Notably, specific strategies translate environmental feedback into textual encapsulations, bolstering agents’ contextual grasp for future engagements [138].

- **Compressing memories with vectors or data structures.** Using appropriate data structures, LLM-based agents can store historical interactions more efficiently. Notably, several methodologies lean on embedding vectors for memory sections, plans, or dialogue histories [90, 137, 139, 140]. Another approach translates sentences into triplet configurations [141], while some perceive memory as a unique data object, fostering varied interactions [142]. Furthermore, ChatDB [143] and DB-GPT [144] opt for SQL databases, which allows for data manipulation via SQL queries.

Methods for memory retrieval. When an agent interacts with its environment, it is imperative to retrieve the most appropriate content from its memory. This ensures that the agent accesses relevant and accurate information to make decision [137, 140]. A representative approach in automated retrieval considers three metrics: recency, relevance, and importance. The memory score is determined as a weighted combination of these metrics, with memories having the highest scores being prioritized in the model’s context [16]. Some work introduces the concept of interactive memory objects, representing dialogue history that can be moved, edited, deleted, or combined through summarization [142]. Users can view and manipulate these objects, influencing how the agent perceives the dialogue. Similarly, other studies allow for memory operations like deletion based on specific commands provided by users [143]. Such methods ensure that the memory content aligns closely with user expectations.

3.1.3 *Reasoning and planning*

Reasoning. Reasoning, crucial in human intellectual activities for problem-solving, decision-making, and critical analysis, is underpinned by evidence and logic [145–147]. Deductive, inductive, and abductive reasoning are key forms recognized in these endeavors [148]. For LLM-based agents, like humans, reasoning capacity is crucial for addressing complex tasks [19].

For LLMs, the reasoning ability is considered an emergent ability, i.e., such an ability emerges once the language model reaches a certain scale in size [20, 149]. Notably, the CoT method [76, 77] has been

demonstrated to elicit the reasoning capacities of LLMs by guiding them to generate rationales before outputting answers. Other techniques like self-consistency [78], self-polish [80], self-refine [150], and selection-inference [151] also enhance LLM performance. To further improve the reasoning ability of LLMs or LLM-based agents, current research primarily explores supervised fine-tuning (SFT) methods [152] and interactive training approaches [153, 154].

Planning. Planning is a key strategy employed by humans when facing complex tasks. For humans, planning helps organize thoughts, set objectives, and determine the steps to achieve those objectives [155–157]. Similar to humans, the ability to plan is crucial for agents, and central to this planning module is the capacity for reasoning [158–160]. This offers a structured thought process for agents based on LLMs. Through reasoning, agents break down complex tasks into manageable sub-tasks, devising appropriate plans for each [161, 162]. Moreover, as tasks progress, agents can employ introspection to modify their plans, ensuring they align better with real-world circumstances, facilitating adaptive and successful task execution. Typically, planning comprises two stages: plan formulation and plan reflection.

- **Plan formulation.** During plan formulation, agents generally decompose an overarching task into numerous sub-tasks, and various approaches have been proposed in this phase. Notably, some studies advocated for LLM-based agents to decompose problems comprehensively in a single step, formulating a complete plan at once and then executing it sequentially [79, 163–165]. In contrast, other studies like the CoT-series employ an adaptive strategy, where they plan and address sub-tasks one at a time, allowing for more fluidity in handling intricate tasks in their entirety [76, 77, 166]. Additionally, some methods emphasize hierarchical planning [167, 168], while others underscore a strategy in which final plans are derived from reasoning steps structured in a tree-like format. The latter approach argues that agents should assess all possible paths before finalizing a plan [78, 169–171]. While LLM-based agents demonstrate a broad scope of general knowledge, they can occasionally face challenges when tasked with situations that require expertise knowledge. Enhancing these agents by integrating them with planners of specific domains has shown to yield better performance [109, 115, 172, 173].

- **Plan reflection.** After formulating a plan, it is imperative to reflect upon and evaluate its merits. LLM-based agents employ internal feedback mechanisms, often drawing insights from pre-existing models, to hone and enhance their strategies and planning approaches [138, 150, 174, 175]. To better align with human values and preferences, agents actively engage with humans, allowing them to rectify misunderstandings and assimilate this tailored feedback into their planning methodology [89, 176, 177]. Furthermore, they could gather feedback from tangible or virtual surroundings, such as cues from task accomplishments or post-action observations, aiding them in revising and refining their plans [72, 82, 178–180].

3.1.4 *Transferability and generalization*

The remarkable nature of the human brain is largely attributed to its high degree of plasticity and adaptability. It can continuously adjust its structure in response to external stimuli and internal needs, thereby adapting to different environments and tasks. These years, plenty of research indicates that pre-trained models can learn universal language representations [181–183], and with only a small amount of data for fine-tuning, it can demonstrate excellent performance in downstream tasks [184]. There is no need to train new models from scratch, which saves a lot of computation resources. However, models trained through this task-specific fine-tuning, usually lack versatility and generalizability to other tasks. Instead of merely functioning as a static knowledge repository, LLM-based agents exhibit dynamic learning ability which enables them to adapt to novel tasks swiftly and robustly [18, 86, 87].

Unseen task generalization. Studies show that instruction-tuned LLMs exhibit zero-shot generalization without the need for task-specific fine-tuning [18, 19, 86–88]. Notably, LLMs can complete unfamiliar tasks by following the instructions based on their own understanding. One of the implementations is multi-task learning, for example, FLAN [86] finetunes language models on a collection of tasks described via instructions, and T0 [87] introduces a unified framework that converts every language problem into a text-to-text format. Despite being purely a language model, GPT-4 [19] demonstrates remarkable capabilities in a variety of domains and tasks, including abstraction, coding, mathematics, medicine, law, and others [24]. Promisingly, such generalization capability can be further enhanced by scaling up the model size and the quantity or diversity of training instructions [75, 185].

In-context learning. Numerous studies indicate that LLMs can perform a variety of complex tasks through in-context learning (ICL), which involves the models' ability to learn from a few examples within a given context [186]. Few-shot in-context learning enhances the performance by concatenating the

original input with several complete examples as prompts to enrich the context [69]. The key idea of ICL is learning from analogy, akin to the learning process of humans [187]. Besides, since the prompts are written in natural language, the interaction is interpretable and changeable, making it easier to incorporate human knowledge [76, 188]. Unlike the supervised learning process, ICL does not involve fine-tuning or parameter updates, greatly reducing computation costs for adapting the models to new tasks. Beyond texts, researchers also explore the potential ICL capabilities in different multimodal tasks [189–194], making it possible for agents to be applied to large-scale real-world tasks.

3.2 Perception

Both humans and animals rely on sensory organs like eyes and ears to gather information from their surroundings. These perceptual inputs are converted into neural signals and sent to the brain for processing [195, 196], allowing humans to perceive and interact with the world. Similarly, it is crucial for LLM-based agents to receive information from various sources and modalities. This expanded perceptual space helps agents better understand their environment, make informed decisions, and excel in a broader range of tasks, making it an essential development direction. Agent handles this information to the Brain module for processing through the perception module.

In this section, we introduce how to enable LLM-based agents to acquire multimodal perception capabilities, encompassing textual (Subsection 3.2.1), visual (Subsection 3.2.2), and auditory inputs (Subsection 3.2.3). We also consider other potential input forms (Subsection 3.2.4) such as tactile feedback, gestures, and 3D maps to enrich the agent’s perception domain and enhance its versatility.

3.2.1 Textual input

Text serves as a conduit for conveying data, information, and knowledge, making text communication one of the most important ways through which humans interact with the world. LLM-based agents already have the fundamental ability to communicate with humans through textual input and output [95]. In a user’s textual input, aside from the explicit content, lie concealed beliefs, desires, and intentions. Understanding implied meanings is crucial for the agent to grasp the potential and underlying intentions of human users, thereby enhancing its communication efficiency and quality with users. However, understanding implied meanings within textual input remains challenging for current LLM-based agents [75, 197]. For example, some studies [113, 198–200] employed reinforcement learning to perceive implied meanings and models feedback to derive rewards. This helps deduce the speaker’s preferences, leading to more personalized and accurate responses from the agent. Additionally, as the agent is designed for use in complex real-world situations, it will inevitably encounter many entirely novel tasks. Understanding text instructions for unknown tasks places higher demands on the agent’s text perception abilities. As described in Subsection 3.1.4, an LLM that has undergone instruction tuning [86] can exhibit remarkable zero-shot instruction understanding and generalization abilities, eliminating the need for task-specific fine-tuning.

3.2.2 Visual input

Although LLMs excel in language comprehension [19, 201] and multi-turn conversations [202], they inherently lack visual perception and can only understand discrete textual content. Visual input usually contains a wealth of information about the world, including properties of objects, spatial relationships, scene layouts, and more in the agent’s surroundings. Therefore, integrating visual information with data from other modalities can offer the agent a broader context and a more precise understanding [101], deepening the agent’s perception of the environment.

To help the agent understand the information contained within images, a straightforward approach is to generate corresponding text descriptions for image inputs, known as image captioning [203–207]. Captions can be directly linked with standard text instructions and fed into LLM-based agents. This approach is highly interpretable and does not require additional training for caption generation, which can save a significant number of computational resources. However, caption generation is a low-bandwidth method [101, 208], and it may lose a lot of potential information during the conversion process. Furthermore, the agent’s focus on images may introduce biases.

The second approach to endowing an LLM-based agent with visual understanding capabilities is modality fusion, where a visual encoder is integrated with an LLM at the embedding level [209, 210]. Freezing

one or both of them during training is a widely adopted paradigm that achieves a balance between training resources and model performance [211]. However, LLMs cannot directly understand the output of a visual encoder, so it is necessary to convert the image encoding into embeddings that LLMs can comprehend. In other words, it involves aligning the visual encoder with the LLM. This usually requires adding an extra learnable interface layer between them. For example, BLIP-2 [211] and InstructBLIP [212] use the querying transformer (Q-Former) module as an intermediate layer between the visual encoder and the LLM [212]. At the same time, some researchers adopt a computationally efficient method using a single projection layer to achieve visual-text alignment, reducing the need for training additional parameters [99, 210, 213]. Moreover, the projection layer can effectively integrate with the learnable interface to adapt the dimensions of its outputs, making them compatible with LLMs [214–217].

3.2.3 Auditory input

Auditory information constitutes a crucial component of world information. When an agent possesses auditory capabilities, it can improve its awareness of interactive content, the surrounding environment, and even potential dangers. While there are numerous well-established models and approaches [218–220] for processing audio as a standalone modality, these models often only excel at specific tasks. Given the excellent tool-using capabilities of LLMs (which will be discussed in detail in Subsection 3.3), a very intuitive idea is that the agent can use LLMs as control hubs, invoking existing toolsets or model repositories in a cascading manner to perceive audio information.

Furthermore, an audio spectrogram can serve as a medium to endow LLM-based agents with auditory capabilities, as it offers a clear representation of how the frequency spectrum of an audio signal evolves over time [221]. For a segment of audio data over a period of time, it can be abstracted into a finite-length audio spectrogram. An audio spectrogram has a 2D representation, which can be visualized as a flat image. Hence, some research [222, 223] efforts aim to migrate perceptual methods from the visual domain to audio. Audio spectrogram transformer (AST) [222] employs a Transformer architecture similar to ViT to process audio spectrogram images. By segmenting the audio spectrogram into patches, it achieves effective encoding of audio information.

3.2.4 Other input

LLM-based agents are expected to be equipped with richer perception modules in the future. They could perceive and understand diverse modalities in the real world, much like humans. For example, agents could have unique touch and smell organs, allowing them to gather more detailed information when interacting with objects. At the same time, agents can also be equipped with hardware devices, such as GPS, Lidar, and other sensors, having a clearer sense of the spatial position, temperature, and brightness and taking environment-aware actions.

3.3 Action

After humans perceive their environment, their brains integrate, analyze, and reason to make decisions. Subsequently, they employ their nervous systems to control their bodies, enabling adaptive or creative actions in response to the environment, such as engaging in conversation, evading obstacles, or starting a fire. When an agent possesses a brain-like structure with capabilities of knowledge, memory, reasoning, planning, and generalization, as well as multimodal perception, it is also expected to possess a diverse range of actions akin to humans to respond to its surrounding environment. In constructing such an agent, the action module receives action sequences sent by the brain module and carries out actions to interact with the environment.

This section begins with textual output (Subsection 3.3.1), which is the inherent capability of LLM-based agents. Next we talk about the tool-using capability of LLM-based agents (Subsection 3.3.2), which has proved effective in enhancing their versatility and expertise. Finally, we discuss equipping the LLM-based agent with embodied action to facilitate its grounding in the physical world (Subsection 3.3.3).

3.3.1 Textual output

Recently, the rise and development of Transformer-based generative LLMs have endowed LLM-based agents with inherent language generation capabilities [19, 224]. The text quality they generate excels in

various aspects such as fluency, relevance, diversity, controllability [112, 225–227]. Consequently, LLM-based agents can be exceptionally strong language generators.

3.3.2 Tool using

Tools are extensions of the capabilities of humans. When faced with complex tasks, humans employ tools to simplify task-solving and enhance efficiency, freeing time, and resources. Similarly, agents have the potential to accomplish complex tasks more efficiently and with higher quality if they also learn to use and utilize tools [75]. LLM-based agents have limitations because of hallucination [228], the lack of training data and tuning for specific fields [229], untransparent decision-making process [230] and susceptibility to adversarial attacks [231]. Fortunately, specialized tools enable LLM-based agents to enhance their expertise, adapt domain knowledge, and be more suitable for domain-specific needs in a pluggable form. They also exhibit stronger interpretability and robustness. LLM-based agents not only require the use of tools, but are also well-suited for tool integration. LLMs show remarkable reasoning and decision-making abilities in complex interactive environments [78] and significant potential in intent understanding and other aspects [19, 232–234]. These make LLM-based agents proficient tool users.

Learning to use tools. Leveraging the powerful zero-shot and few-shot learning abilities of LLMs [69, 235], agents can acquire knowledge about tools by utilizing zero-shot prompts that contain descriptions of tool functionalities and parameters, or few-shot prompts that provide demonstrations of specific tool usage scenarios and corresponding methods [73, 236]. These learning approaches parallel human methods of learning by consulting tool manuals or observing others using tools [75]. Besides configurations and demonstrations, agents can also learn from feedback received from both the environment and humans [18, 237, 238]. Environmental feedback encompasses result feedback on whether actions have successfully completed the task and intermediate feedback that captures changes in the environmental state caused by actions; human feedback comprises explicit evaluations and implicit behaviors, such as clicking on links [75].

It is crucial to improve the agent’s generalization ability in tool usage to handle complicated scenarios as well as continuously evolving new tools. To accomplish this, agents need to grasp the common principles or patterns in tool usage strategies, which can potentially be achieved through meta-tool learning [239]. In addition, techniques like curriculum learning [240] can enhance the agent’s understanding of relationships between simple and complex tools, such as how complex tools are built on simpler ones and allow agents to effectively discern nuances across various application scenarios and transfer previously learned knowledge to new tools [75]. Previous studies showed that LLM-based agents not only have the ability to generalize the use of existing tools, but can also make new tools they need by generating executable programs, or integrating existing tools into more powerful ones [75, 241, 242].

Tools expanding the action space of LLM-based agents. With the help of tools, agents can utilize various external resources such as web applications and other LMs during the reasoning and planning phase [73]. This process can provide information with high expertise, reliability, diversity, and quality for LLM-based agents, facilitating their decision-making and action. For example, search-based tools can improve the scope and quality of the knowledge accessible to the agents with the aid of external databases, knowledge graphs, and web pages, while domain-specific tools can enhance an agent’s expertise in the corresponding field [243, 244]. Some researchers have already developed LLM-based controllers that generate SQL statements to query databases, or to convert user queries into search requests and use search engines to obtain the desired results [71, 143]. What is more, LLM-based agents can use scientific tools to execute tasks like organic synthesis in chemistry, or interface with Python interpreters to enhance their performance on intricate mathematical computation tasks [245, 246].

Although the tools mentioned before enhance the capabilities of agents, the medium of interaction with the environment remains text-based. However, tools are designed to expand the functionality of language models, and their outputs are not limited to text. Tools for non-textual output can diversify the modalities of agent actions, thereby expanding the application scenarios of LLM-based agents [163, 247]. For example, image processing and generation can be accomplished by an agent that draws on a visual model [248].

3.3.3 Embodied action

In the pursuit of AGI, the embodied agent is considered a pivotal paradigm. The embodiment hypothesis [249] draws inspiration from the human intelligence development process, posing that an agent’s

intelligence arises from continuous interaction and feedback with the environment rather than relying solely on well-curated textbooks [250]. Similarly, people anticipate that LLM-based agents should be capable of actively perceiving, comprehending, and interacting with physical environments, making decisions, and generating intended behaviors to modify the environment based on LLM's internal knowledge. We collectively term these as embodied actions, which enable agents' ability to interact with and comprehend the world in a manner closely resembling human behavior.

The potential of LLM-based agents for embodied actions. Despite the extensive success of RL-based embodiment [50, 51, 251], it does have certain limitations in some aspects. For example, RL algorithms face limitations in terms of cost efficiency, generalization, and complex problem reasoning due to challenges in modeling the dynamic environment and the reliance on reward signal representations [252]. Recent studies have indicated that leveraging the rich internal knowledge acquired during the pre-training of LLMs can effectively alleviate these issues [101, 171, 178, 253].

- **Cost efficiency.** Some on-policy algorithms struggle with sample efficiency as they require iteratively sampled data for policy updates while gathering enough embodied data for high-performance training is costly. The constraint is also found in some end-to-end models [254–256]. By leveraging the intrinsic knowledge from LLMs, agents like PaLM-E [101] jointly train robotic data with general visual-language data to achieve significant transfer ability in embodied tasks while also showcasing that geometric input representations can improve training data efficiency.

- **Embodied action generalization.** As discussed in Subsection 3.1.4, an agent's competence should extend beyond specific tasks. Different from the majority of RL algorithms [82, 257–259], fine-tuned by tasks in diverse forms and types, LLMs have showcased remarkable cross-task generalization capabilities [260, 261]. Further, natural language serves both as a means to interact with the environment and as a medium for transferring foundational skills to new tasks [262]. SayCan [163] decomposes task instructions presented in prompts using LLMs into corresponding skill commands, but in partially observable environments, limited prior skills often do not achieve satisfactory performance [82]. To address this, Voyager [177] introduces the skill library component to continuously collect novel self-verified skills, which allows for the agent's lifelong learning capabilities.

- **Embodied action planning.** Planning is a pivotal strategy for agents when addressing complex problems. In traditional hierarchical reinforcement learning (HRL) methods, the high-level policy constraints sub-goals for the low-level policy which produces appropriate action signals [263–265]. Similar to the role of high-level policies, LLMs with emerging reasoning abilities [20] can be seamlessly applied to complex tasks in a zero-shot or few-shot manner [76, 78–80]. In addition, external feedback from the environment can further enhance LLM-based agents' planning performance. Some studies [72, 81, 82, 266] dynamically generated, maintained, and adjusted high-level action plans in order to minimize dependency on prior knowledge, thereby grounding the plan. Such feedback can also come from models and humans, which can usually be referred to as the critics, assessing task completion based on the current state and task prompts [19, 177].

Embodied actions for LLM-based agents. There are several fundamental embodied actions or tasks for LLM-based agents to master, primarily including observation, manipulation, and navigation.

- **Observation.** Observation plays a crucial role in subsequent embodied actions by which the agent acquires environmental information and updates states. As mentioned in Subsection 3.2, during observation stage, various inputs are ultimately converged into a multimodal signal. A common approach entails a pre-trained vision transformer (ViT) used as the alignment module for text and visual information [101, 102, 267]. In recent times, more researches take audio as a modality for embedded observation. Soundspaces [268] proposes the identification of spatial geometric elements guided by reverberant audio input [265]. Apart from the cascading paradigm [218, 219, 269], audio information encoding can also enhance the seamless integration of audio with other modalities of inputs [222]. Additionally, the agents' observation could be from real-time human linguistic instructions, which helps the agent in acquiring detail information that may not be readily obtained or parsed [177, 270].

- **Manipulation.** Embodied agents' manipulation tasks generally include object rearrangements, tabletop manipulation, and mobile manipulation [17, 101]. The typical case entails the agent executing a sequence of tasks in the kitchen, which includes retrieving items from drawers and handing them to the user [163]. This task involves combining a series of subgoals and maintaining synchronization between the agent's state and such subgoals are of significance [271]. Besides these, AlphaBlock [272] focuses on more challenging manipulation tasks (e.g., making a smiley face using building blocks), which requires

agents to have a more grounded understanding of the instructions. It fine-tunes a multimodal model based on a complex task dataset comprising corresponding multi-step planning and observation pairs to enhance its comprehension of high-level cognitive instructions.

- **Navigation.** In navigation tasks, agents need to dynamically alter their positions, which often involves multi-angle and multi-object observations [17]. Before navigation, it is essential for embodied agents to establish prior internal maps, typically in the form of a topological map, semantic map, and occupancy map [250]. For example, LM-Nav [273] utilizes the VNM [274] to create the topological map, and further leverages the LLM and VLM for analyzing the environment to find the optimal path. Some studies [275, 276] highlighted the importance of spatial representation to obtain the precise localization of targets by leveraging the pre-trained VLM model to combine visual features from images with 3D reconstructions of the physical world [250]. During navigation tasks, the states of the agent are often influenced by its past actions. A memory mechanism is needed to record historical information [277], which is also adopted in Smallville and Voyager [16, 177, 278, 279]. Additionally, audio input is also of great significance for navigation tasks. Previous studies demonstrate a basic framework that includes a dynamic path planner that uses visual and auditory observations along with spatial memories to plan a series of actions [265, 280].

Complex and grounding embodied actions. By integrating actions above, the agent can accomplish more complex tasks, such as embodied question answering, e.g., Is the watermelon in the kitchen larger than the pot? Which one is harder? To address these questions, the agent needs to navigate to the kitchen, observe the sizes of both objects, and then answer the questions through comparison [250]. In terms of control strategies, as previously mentioned, LLM-based agents trained on particular embodied datasets typically generate high-level policy commands to control low-level policies [163] like a robotic transformer [101, 281, 282], which takes images and instructions as inputs and produces control commands for the end effector and robotic arms as well as virtual embodied controllers due to high costs of robotic operators [16, 89, 90, 139, 271, 283, 284]. By utilizing the Mineflayer [285] API, some studies enable cost-effective examination of embodied agents' operations including exploration, planning, and lifelong learning [177]. On the other hand, grounding language to action space poses an obstacle for agents. For example, understanding the linguistic metaphor expressions like "jump down like a cat" requires adequate world knowledge [286]. Summers et al. [287] endeavored to amalgamate text distillation with hindsight experience replay to construct a dataset for training. Nevertheless, additional investigation on grounding embodied action still remains necessary as it plays an increasingly pivotal role across various domains in human life.

4 Agents in practice: harnessing AI for good

The LLM-based agent, as an emerging direction, has gained increasing attention from researchers. Many applications in specific domains and tasks have already been developed, showcasing the powerful and versatile capabilities of agents [288]. As an LLM-based agent, its design objective should always be beneficial to humans, i.e., humans can harness AI for good.

In this section, we provide an overview of current applications of LLM-based agents, aiming to offer a broad perspective for the practical deployment scenarios (see Figure 3). First, we elucidate the diverse application scenarios of single agent, including task-oriented, innovation-oriented, and lifecycle-oriented scenarios (Subsection 4.1). Then, we present the significant coordinating potential of multiple agents: cooperative interaction or adversarial interaction (Subsection 4.2). Finally, we categorize the interactive collaboration between humans and agents into two paradigms and introduce the specific applications (Subsection 4.3).

4.1 General ability of single agent

Currently, there is a vibrant development of application instances of LLM-based agents [289–291]. AutoGPT [95] is one of the ongoing popular projects aiming to achieve a fully autonomous system. Apart from the basic functions of LLMs, the AutoGPT framework also incorporates various external tools and memory management. After users input their customized objectives, they can free their hands and wait for AutoGPT to automatically generate thoughts and perform specific tasks, all without requiring additional user prompts. As shown in Figure 4, we introduce the astonishingly capabilities that the agent exhibits in scenarios where only one single agent is present.



Figure 3 (Color online) Scenarios of LLM-based agent applications: single-agent deployment, multi-agent interaction, and human-agent interaction. A single agent possesses diverse capabilities and demonstrates outstanding task-solving performance in various application orientations. When multiple agents interact, they can achieve advancement through cooperative or adversarial interactions. Furthermore, in human-agent interactions, human feedback can enable agents to perform tasks more efficiently and safely, while agents can also provide better service to humans.

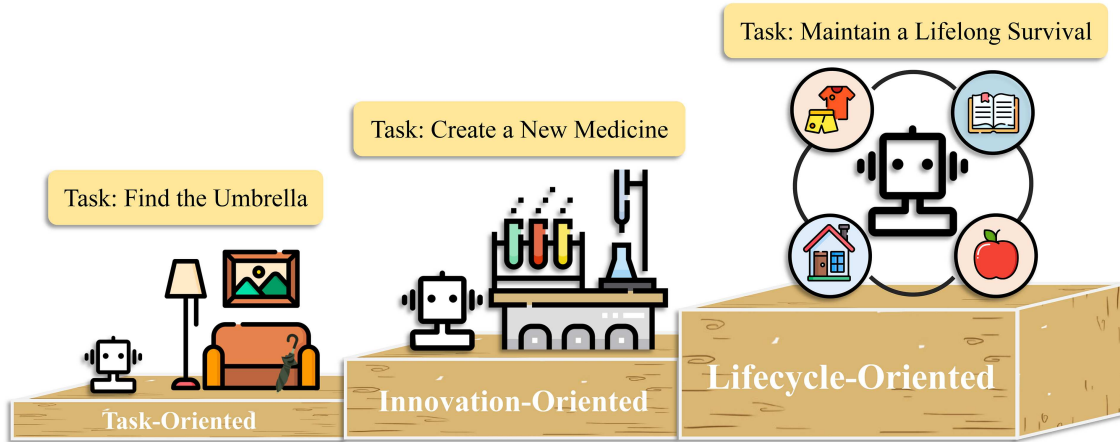


Figure 4 (Color online) Practical applications of the single LLM-based agent in different scenarios. In task-oriented deployment, agents assist human users in solving daily tasks. They need to possess basic instruction comprehension and task decomposition abilities. In innovation-oriented deployment, agents demonstrate the potential for autonomous exploration in scientific domains. In lifecycle-oriented deployment, agents have the ability to continuously explore, learn, and utilize new skills to ensure long-term survival in an open world.

4.1.1 Task-oriented deployment

The LLM-based agents, which can understand human natural language instructions and perform everyday tasks [292], are currently among the most favored and practically valuable agents by users. This is because they have the potential to enhance task efficiency, alleviate user workload, and promote access for a broader user base. In task-oriented deployment, the agent follows high-level instructions from users, undertaking tasks such as goal decomposition [167,171,293,294], sequence planning of sub-goals [167,295], interactive exploration of the environment [165,292,296,297], until the final objective is achieved. Based on task types, we divide these deployment environments into web scenarios and life scenarios, and introduce the specific roles that agents play in them.

In web scenarios. Performing specific tasks on behalf of users in a web scenario is known as the web navigation problem [296]. Agents interpret user instructions, break them down into multiple basic operations, and interact with computers. This often includes web tasks such as filling out forms, online shopping, and sending emails. Agents need to possess the ability to understand instructions within complex web scenarios, adapt to changes (such as noisy text and dynamic HTML web pages), and generalize successful operations [292]. In this way, agents can achieve accessibility and automation when dealing with unseen tasks in the future [298], freeing humans from repeated interactions with computer UIs.

To enable successful interactions between agents and realistic web pages, some researchers [294,299] have started to leverage the powerful HTML reading and understanding abilities of LLMs. By designing prompts, they attempt to make agents understand the entire HTML source code and predict more reasonable next action steps. Mind2Web [300] combines multiple LLMs fine-tuned for HTML, allowing them to summarize verbose HTML code [293] in real-world scenarios and extract valuable information. Furthermore, WebGum [296] empowers agents with visual perception abilities by employing a multimodal corpus containing HTML screenshots. Auto-GUI [301] also proposes a multimodal solution that directly interacts with the interface on mobile devices, thus deepening the comprehensive understanding of web

pages.

In life scenarios. In many daily household tasks, it is essential for agents to understand implicit instructions and apply common-sense knowledge [302]. For an LLM-based agent trained solely on massive amounts of text, tasks that humans take for granted might require multiple trial-and-error attempts [303]. More realistic scenarios often lead to more obscure tasks. For example, the agent should proactively turn it on if it is dark and there is a light in the room. To successfully chop some vegetables in the kitchen, the agent needs to anticipate the possible location of a knife [167].

Can an agent apply the world knowledge embedded in its training data to real interaction scenarios? Huang et al. [171] led the way in exploring this question. They demonstrated that sufficiently large LLMs, with appropriate prompts, can effectively break down high-level tasks into suitable sub-tasks without additional training. However, this static reasoning and planning ability has its potential drawbacks. Actions generated by agents often lack awareness of the dynamic environment around them. For instance, when a user gives the task “clean the room”, the agent might convert it into unfeasible sub-tasks like “call a cleaning service” [304]. As a result, some approaches directly incorporate spatial data and item-location relationships as additional inputs to the model. This allows agents to gain a precise description of their surroundings and plan next actions more effectively [167, 295, 304].

4.1.2 *Innovation-oriented deployment*

The LLM-based agent has demonstrated strong capabilities in performing tasks. However, in a more intellectually demanding field, like cutting-edge science, the potential of agents has not been fully realized yet. This limitation mainly arises from two challenges [305]. On one hand, **the inherent complexity of science poses a significant barrier**. Many domain-specific terms and multi-dimensional structures are difficult to represent using a single text. As a result, their complete attributes cannot be fully encapsulated. On the other hand, **there is a severe lack of suitable training data in scientific domains**, making it difficult for agents to comprehend the entire domain knowledge [306, 307]. If the ability for autonomous exploration could be discovered within the agent, it would bring about beneficial innovation in human technology.

Currently, numerous efforts in various specialized domains aim to overcome this challenge [308–310]. Experts from the computer field make full use of the agent’s powerful code comprehension and debugging abilities [288, 311]. **In the fields of chemistry and materials, researchers equip agents with various general or task-specific tools to better understand domain knowledge**. Agents evolve into scientific assistants, proficient in online research and document analysis to fill data gaps. They also employ robotic APIs for real-world interactions, enabling tasks like material synthesis and mechanism discovery [91, 245, 305].

The potential of LLM-based agents in scientific innovation is evident, yet we do not expect their exploratory abilities to be utilized in applications that could threaten or harm humans. Boiko et al. [91] study the hidden dangers of agents in synthesizing illegal drugs and chemical weapons, indicating that agents could be misled by malicious users in adversarial prompts. This serves as a warning for our future work.

4.1.3 *Lifecycle-oriented deployment*

Building a universally capable agent that can continuously explore, develop new skills, and maintain a long-term life cycle in an open, unknown world is a colossal challenge. This accomplishment is regarded as a pivotal milestone in the field of AGI [271]. Minecraft, as a typical and widely explored simulated survival environment, has become a unique playground for developing and testing the comprehensive ability of an agent. Players typically start by learning the basics, such as mining wood and making crafting tables, before moving on to more complex tasks like fighting against monsters and crafting diamond tools [177]. Minecraft fundamentally reflects the real world, making it conducive for researchers to investigate an agent’s potential to survive in the authentic world.

The survival algorithms of agents in Minecraft can generally be categorized into two types [177]: low-level control and high-level planning. Early efforts mainly focused on reinforcement learning [177, 312] and imitation learning [313], enabling agents to craft some low-level items. With the emergence of LLMs, which demonstrated surprising reasoning and analytical capabilities, agents begin to utilize LLM as a high-level planner to guide simulated survival tasks [271, 314]. Some researchers use LLM to decompose high-level task instructions into a series of sub-goals [315], basic skill sequences [314], or fundamental keyboard/mouse operations [315], gradually assisting agents in exploring the open world.

Voyager [177], drawing inspiration from AutoGPT [95], became the first LLM-based embodied lifelong learning agent in Minecraft. It introduces a skill library for storing and retrieving complex action-executable code, along with an iterative prompt mechanism that incorporates environmental feedback and error correction. This enables the agent to autonomously explore and adapt to unknown environments without human intervention. An AI agent capable of autonomously learning and mastering the entire real-world techniques may not be as distant as once thought [315].

4.2 Coordinating potential of multiple agents

Motivation and background. Although LLM-based agents excel in text understanding and generation, they inherently function as isolated entities [316]. A key limitation of the single-agent application paradigm is their ability to collaborate with other agents and learn from social interactions. This limitation restricts their capacity to benefit from multi-turn feedback, which could otherwise improve their performance [21].

As early as 1986, Minsky [317] made a forward-looking prediction. In his book *The Society of Mind*, he introduced a novel theory of intelligence, suggesting that intelligence emerges from the interactions of many smaller agents with specific functions. For instance, certain agents might be responsible for pattern recognition, while others might handle decision-making or generate solutions. This idea has been put into concrete practice with the rise of distributed artificial intelligence [318]. Multi-agent systems (MAS) [3], as one of the primary research domains, focus on how a group of agents can effectively coordinate and collaborate to solve problems. Some specialized communication languages, like KQML [319], were designed early on to support message transmission and knowledge sharing among agents. In the 21st century, integrating reinforcement learning algorithms (such as Q-learning) with deep learning has become a prominent technique for developing MAS that operate in complex environments [320]. Nowadays, the construction approach based on LLMs is beginning to demonstrate remarkable potential. The natural language communication between agents has become more elegant and easily comprehensible to humans, resulting in a significant leap in interaction efficiency.

Potential advantages. Specifically, an LLM-based multi-agent system can offer several advantages. Just as Smith clearly stated in *The Wealth of Nations* [321], “The greatest improvements in the productive powers of labor, and most of the skill, dexterity, and judgment with which it is directed or applied, seem to be results of the division of labor.” Based on the principle of division of labor, a single agent equipped with specialized skills and domain knowledge can engage in specific tasks. On the one hand, agents’ skills in handling specific tasks are increasingly refined through the division of labor. On the other hand, decomposing complex tasks into multiple subtasks can eliminate the time spent switching between different processes. In the end, efficient division of labor among multiple agents can accomplish a significantly greater workload than when there is no specialization, substantially improving the overall system’s efficiency and output quality.

In Subsection 4.1, we have provided a comprehensive introduction to the versatile abilities of LLM-based agents. In this subsection, we focus on exploring the ways agents interact with each other in a multi-agent environment. Based on current research, these interactions can be broadly categorized as follows: cooperative interaction and adversarial interaction (see Figure 5).

4.2.1 Cooperative interaction for complementarity

Cooperative multi-agent systems are the most widely deployed pattern in practical usage. Within such systems, individual agent assesses the needs and capabilities of other agents and actively seeks collaborative actions and information sharing with them [89]. This approach brings forth numerous potential benefits, including enhanced task efficiency, collective decision improvement, and the resolution of complex real-world problems that one single agent cannot solve independently. We introduce and categorize existing cooperative multi-agent applications into two types: disordered cooperation and ordered cooperation.

Disordered cooperation. When three or more agents are present within a system, each agent is free to express their perspectives and opinions openly. They can provide feedback and suggestions for modifying responses related to the task at hand [322]. This entire discussion process is uncontrolled, lacking any specific sequence. We refer to this kind of multi-agent cooperation as disordered cooperation.

ChatLLM network [323] is an exemplary representative of this concept. It emulates the forward and backward propagation process within a neural network, treating each agent as an individual node. Agents

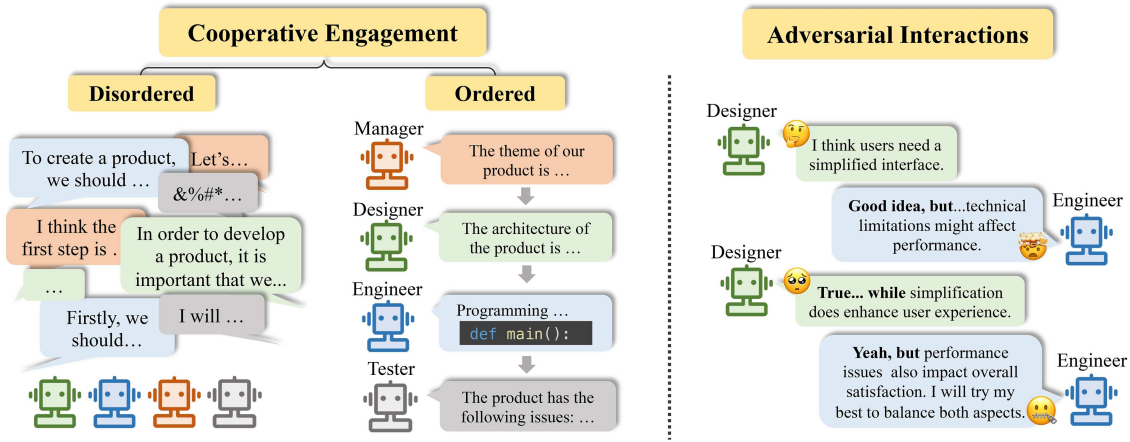


Figure 5 (Color online) Interaction scenarios for multiple LLM-based agents. In cooperative interaction, agents collaborate in either a disordered or ordered manner to achieve shared objectives. In adversarial interaction, agents compete in a tit-for-tat fashion to enhance their respective performance.

in the subsequent layer need to process inputs from all the preceding agents and propagate forward. One potential solution is introducing a dedicated coordinating agent, responsible for integrating and organizing responses from all agents, thus updating the final answer [324]. However, consolidating a large amount of feedback data and extracting valuable insights pose a significant challenge for the coordinating agent. Furthermore, majority voting can also serve as an effective approach to making appropriate decisions [325].

Ordered cooperation. When agents in the system adhere to specific rules, for instance, expressing their opinions one by one in a sequential manner, downstream agents only need to focus on the outputs from upstream. This leads to a significant improvement in task completion efficiency. The entire discussion process is highly organized and ordered. We term this kind of multi-agent cooperation as ordered cooperation. It is worth noting that systems with only two agents, essentially engaging in a conversational manner, also fall under the category of ordered cooperation.

CAMEL [89] stands as a successful implementation of a dual-agent cooperative system. Within a role-playing communication framework, agents take on the roles of users (giving instructions) and assistants (providing specific solutions). Through multi-turn dialogues, these agents autonomously collaborate to fulfill user instructions [326]. On the other hand, Talebirad et al. [316] were among the first to systematically introduce a comprehensive LLM-based multi-agent collaboration framework. This paradigm aims to harness the strengths of each individual agent and foster cooperative relationships among them. Many applications of multi-agent cooperation have successfully been built upon this foundation [21, 327–330]. To promote more efficient collaboration, researchers hope that agents can learn from successful human cooperation examples [90]. Wang et al. [331] advocated for the integration of evolutionary game theory and artificial intelligence to advance the mathematics of multi-agent learning systems. MetaGPT [332] draws inspiration from the classic waterfall model in software development. By encoding advanced human management experience into agent prompts, collaboration among multiple agents becomes more structured.

However, during MetaGPT's practical exploration, a potential threat to multi-agent cooperation has been identified. Without setting corresponding rules, frequent interactions among multiple agents can amplify minor hallucinations indefinitely [332]. For example, in software development, issues like incomplete functions, missing dependencies, and bugs that are imperceptible to the human eye may arise. Introducing techniques like cross-validation [90] or timely external feedback could have a positive impact on the quality of agent outputs.

4.2.2 Adversarial interaction for advancement

Traditionally, cooperative methods have been extensively explored in multi-agent systems. However, researchers increasingly recognize that introducing concepts from game theory [333, 334] into systems can lead to more robust and efficient behaviors. In competitive environments, agents can swiftly adjust strategies through dynamic interactions, striving to select the most advantageous or rational actions in response to changes caused by other agents [50, 335]. AlphaGo Zero [336], for instance, is an agent for Go

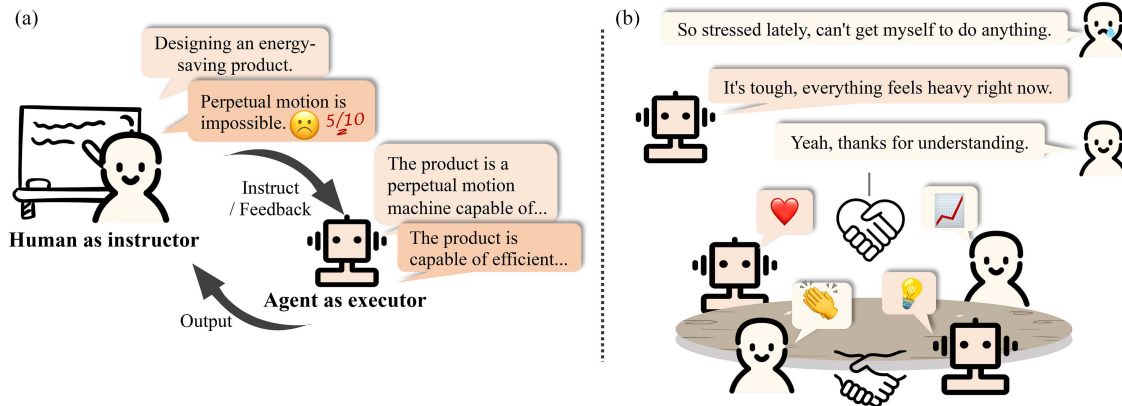


Figure 6 (Color online) Two paradigms of human-agent interaction. In the instructor-executor paradigm (a), humans provide instructions or feedback, while agents act as executors. In the equal partnership paradigm (b), agents are human-like, able to engage in empathetic conversation and participate in collaborative tasks with humans.

that achieved significant breakthroughs through a process of self-play. Similarly, within LLM-based multi-agent systems, fostering change among agents can naturally occur through competition, argumentation, and debate [337, 338]. By abandoning rigid beliefs and engaging in thoughtful reflection, adversarial interaction enhances the quality of responses.

Researchers first delve into the fundamental debating abilities of LLM-based agents [114, 339, 340]. Findings demonstrate that when multiple agents express their arguments in the state of “tit for tat”, one agent can receive substantial external feedback from other agents, thereby correcting its distorted thoughts [93]. Consequently, multi-agent adversarial systems are suitable in scenarios requiring high-quality responses and accurate decision-making. For example, Du et al. [92] introduced the concept of debate, endowing agents with responses from fellow peers. When these responses diverge from an agent’s own judgments, a “mental” argumentation occurs, leading to refined solutions.

The performance of the multi-agent adversarial system has shown considerable promise. However, the system is essentially dependent on the strength of LLMs and faces several basic challenges: (1) With a prolonged debate, LLM’s limited context cannot process the entire input. (2) In a multi-agent environment, computational overhead significantly increases. (3) Multi-agent negotiation may converge to an incorrect consensus, and all agents are firmly convinced of its accuracy [92]. The development of MAS is still far from being mature and feasible. Introducing human guides when appropriate to compensate for agents’ shortcomings is a good choice to promote the further advancements of agents.

4.3 Interactive engagement between human and agent

Human-agent interaction, as the name suggests, involves agents collaborating with humans to accomplish tasks. Throughout the interaction, humans play a pivotal role by offering guidance or by regulating the safety, legality, and ethical conduct of agents [341, 342]. This is particularly crucial in specialized domains, such as medicine where data privacy concerns exist [343]. In such cases, human involvement can serve as a valuable means to compensate for the lack of data, thereby facilitating smoother and more secure collaborative processes. The interaction between humans and agents can be classified into two paradigms (see Figure 6). (1) Unequal interaction (i.e., instructor-executor paradigm), humans serve as issuers of instructions, while agents act as executors, participating as assistants to humans in collaboration. (2) Equal interaction (i.e., equal partnership paradigm): agents reach the level of humans, participating on an equal footing with humans in interaction.

4.3.1 Instructor-executor paradigm

The simplest approach involves human guidance throughout the process: humans provide specific instructions directly, while the agents’ role is to understand natural language commands from humans and translate them into corresponding actions [344–346]. In Subsection 4.1, we have presented the scenario where agents solve single-step problems or receive high-level instructions from humans. In addition, some studies have shown that breaking down instructions into step-by-step formats can improve the cross-task generalization capabilities of language models [347]. Considering the interactive nature of language, in

this section, we assume that the dialogue between humans and agents is also interactive: the agent responds to each human instruction, refining its action through alternating iterations to ultimately meet human requirements [177, 348]. While this approach does achieve the goal of human-agent interaction, it requires a substantial amount of human effort and, in certain tasks, might even necessitate a high level of expertise. To alleviate this issue, the agent can be empowered to autonomously accomplish tasks, while humans only need to provide feedback in certain circumstances. Here, we roughly categorize feedback into two types: quantitative feedback and qualitative feedback.

Quantitative feedback. The forms of quantitative feedback mainly include absolute evaluations like binary scores and ratings, as well as relative scores. Binary feedback refers to the positive and negative evaluations provided by humans, which agents utilize to enhance their self-optimization [349–353]. Comprising only two categories, this type of user feedback is often easy to collect, but sometimes it may oversimplify user intent by neglecting potential intermediate scenarios. To showcase these intermediate scenarios, researchers attempt to expand from binary feedback to rating feedback, which involves categorizing into more fine-grained levels. However, the results of Kreutzer et al. [354] suggest that such multi-level artificial ratings method might be inefficient or less reliable. Furthermore, agents can learn human preference from comparative scores like multiple choice [355, 356].

Qualitative feedback. Text feedback is usually offered in natural language: humans provide advice on how to modify outputs generated by agents, and the agents then incorporate these suggestions to refine their subsequent outputs [357, 358]. For agents without multimodal perception capabilities, humans can act as critics, offering visual critiques [177]. Additionally, agents can utilize a memory module to store feedback for future reuse [359]. In Scheurer et al. [360], humans give feedback on the initial output generated by agents, prompting the agents to formulate various improvement proposals. The agents then discern and adopt the most suitable proposal, harmonizing with the human feedback. Compared to quantitative feedback, this approach can better convey human intention, but it might be more challenging for the agents to comprehend. Xu et al. [361] compared various types of feedback and observed that combining multiple types of feedback can yield better results. Of course, the collaborative nature of human-agent interaction also allows humans to directly improve the agents' output by modifying intermediate links [176, 362] or adjusting the conversation content [363]. In some studies, agents can autonomously judge whether the conversation is proceeding smoothly and seeking feedback when encountering errors [364, 365]. Humans can also choose to participate in feedback at any time, guiding the agent's learning in the right direction [366].

Currently, LLM-based agents serving as human assistants hold tremendous potential across various domains. In education, for instance, the robot Dona [367] supports multimodal interactions to assist students with registration, and Gvirsman et al. [368] contributed to early childhood education by achieving multifaceted interactions. Agents can also aid in human understanding and utilization of mathematics [369]. In the field of medicine, some medical agents have been proposed, showing enormous potential in terms of diagnosis assistance, consultations, and more [370, 371]. Especially in mental health, research has shown that agents can lead to increased accessibility due to benefits such as reduced cost and anonymity compared to face-to-face treatment [372]. Leveraging such advantages, agents have found widespread applications. Ali et al. [373] designed LISSA for online communication with adolescents on the autism spectrum, analyzing their speech and facial expressions in real-time. Hsu et al. [374] built contextualized language generation approaches to provide tailored assistance for users who seek support on diverse topics ranging from relationship stress to anxiety. Besides, in other industries like business, a good agent can provide automated services, thereby effectively reducing labor costs [375]. Amidst the pursuit of AGI, efforts are directed towards creating agents that can function as universal assistants in real-life scenarios [376].

4.3.2 *Equal partnership paradigm*

Empathetic communicator. With the rapid development of AI, conversational agents have garnered extensive attention in various forms [377] and in various scenarios [378–380]. Although it is intuitive that agents themselves do not possess emotions, can we enable them to exhibit emotions and thereby bridge the gap between agents and humans? Therefore, a plethora of research endeavors have embarked on delving into the empathetic capacities of agents. This endeavor seeks to infuse a human touch into these agents, enabling them to detect sentiments and emotions from human expressions, ultimately crafting emotionally resonant dialogues [381–385]. Apart from generating emotionally charged language, agents

can dynamically adjust their emotional states and display them through facial expressions and voice [386]. These studies, viewing agents as empathetic communicators, not only enhance user satisfaction but also make significant progress in fields like healthcare [373, 374, 387] and business marketing [388]. Unlike simple rule-based conversation agents, agents with empathetic capacities can tailor their interactions to meet users' emotional needs [389].

Human-level participant. Furthermore, we hope that agents can be involved in the normal lives of humans, cooperating with humans to complete tasks from a human-level perspective. In many real-world applications, adversarial games play a pivotal role in facilitating highly efficient and effective decision-making processes [390]. While agents have already excelled in pure competitive environments like chess [335], Go [50], and poker [391], LLM-based agents go further by devising unified cooperative strategies through effective negotiation and collaboration in more complex scenarios [392–395]. Beyond games, LLM-based agents demonstrate human-level capabilities in strategy formulation, negotiation, and other interaction-based tasks. They can collaborate with humans, determining shared knowledge, identifying relevant information, posing questions, and reasoning to complete tasks like allocation, planning, and scheduling [396]. Furthermore, LLM-based agents possess persuasive abilities [397], dynamically influencing human viewpoints in various interactive scenarios [398].

In summary, the goal of the field of human-agent interaction is to learn and understand humans, and ultimately enable efficient and secure interactions between humans and agents. Currently, significant breakthroughs have been achieved in terms of usability in this field. In the future, LLM-based agents are expected to provide better assistance to humans in accomplishing more complex tasks in various domains.

5 Agent society: from individuality to sociality

Over an extended duration, researchers and practitioners have envisioned an interactive artificial society wherein human behavior can be performed through trustworthy agents [399]. From sandbox games such as *The Sims* to the concept of Metaverse, we can see how “simulated society” is defined in people’s minds: the environment and the individuals interacting in it. Behind each individual can be a piece of program, a real human, or an LLM-based agent as described in the previous sections [16, 400, 401]. Then, the interaction between individuals also contributes to the emergence of sociality.

In this section, we first introduce the behaviors and personalities of LLM-based agents, tracing their development from individuality to sociality (Subsection 5.1). Subsequently, we introduce a general categorization of the diverse environments for agents to perform their behaviors and engage in interactions (Subsection 5.2). Finally, we discuss how the agent society works, what insights people can get from it, and the risks we need to be aware of (Subsection 5.3).

5.1 Behavior and personality of LLM-based agents

As noted by sociologists, individuals can be analyzed in terms of both external and internal dimensions [402]. The external deals with observable behaviors, while the internal relates to dispositions, values, and feelings. As shown in Figure 7, this framework offers a perspective on emergent behavior and personality in LLM-based agents. Externally, we can observe the sociological behaviors of agents (Subsection 5.1.1), including how agents act individually and interact with their environment. Internally, agents may exhibit intricate aspects of the personality (Subsection 5.1.2), such as cognition, emotion, and character, that shape their behavioral responses.

5.1.1 Social behavior

As Troitzsch et al. [403] stated, the agent society represents a complex system comprising individual and group social activities. Recently, LLM-based agents have exhibited spontaneous social behaviors in an environment [404].

Foundational individual behaviors. Individual behaviors arise through internal cognitive processes and external environmental factors. These behaviors form the basis of how agents operate and develop as individuals within society. They can be classified into three dimensions: (1) Input behaviors refer to the absorption of information from the surroundings. This includes perceiving sensory stimuli [101] and storing them as memories [138]. (2) Internalizing behaviors involve inward cognitive processing. This category encompasses activities such as planning [109], reasoning [76], reflection [72],

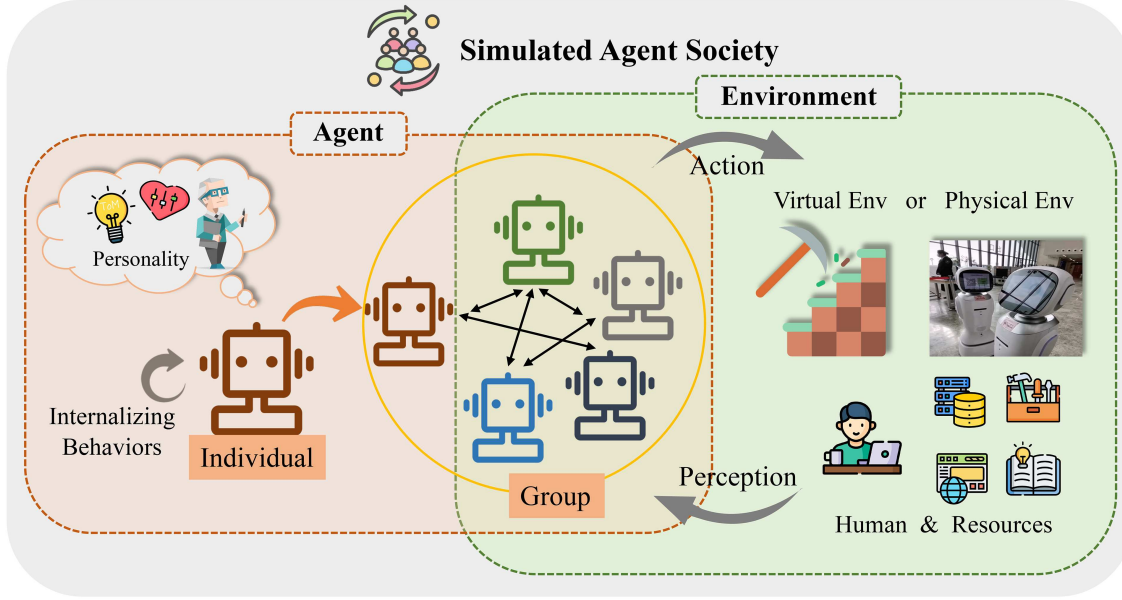


Figure 7 (Color online) Overview of simulated agent society. The whole framework is divided into two parts: the agent and the environment. We can observe in this figure the following. (1) (Left) At the individual level, an agent exhibits internalizing behaviors like planning, reasoning, and reflection. It also displays intrinsic personality traits involving cognition, emotion, and character. (2) (Mid) An agent and other agents can form groups and exhibit group behaviors, such as cooperation. (3) (Right) The environment, whether virtual or physical, contains human actors and all available resources. For a single agent, other agents are also part of the environment. (4) The agents have the ability to interact with the environment via perception and action.

and knowledge precipitation [89,332]. These introspective processes are essential for maturity and self-improvement. (3) Output behaviors include outward actions ranging from object manipulation [101] to structure construction [177]. Moreover, agents express opinions and broadcast information to interact with others [332,405].

Dynamic group behaviors. A group is a gathering of two or more individuals within a defined social context [406]. The attributes of a group evolve due to member interactions and environmental influences. This flexibility gives rise to group behaviors, each with an impact on the larger group. The categories of group behaviors include the following. (1) Positive group behaviors foster collective well-being [16, 90, 322, 327, 328, 340]. An example is cooperation, which is achieved through volunteer behaviors [330], brainstorming discussions [327, 340], and project management [332]. (2) Neutral group behaviors are observed in LLM-based agents. LLMs are designed to be “helpful, honest, and harmless” [407] and this alignment with neutral values [408] leads to conformity behaviors including mimicry and spectating. (3) Negative group behaviors undermine the effectiveness of a group. Recent studies have revealed that agents may exhibit confrontational actions [404] and even resort to destructive behaviors, such as destroying other agents or the environment in pursuit of certain gains [330].

5.1.2 Personality

Recent advances show that LLMs and LLM-based agents exhibit a form of personality through interactions with the group and the environment [409–411]. The widely accepted definition of personality includes cognitive, emotional, and character traits [412].

Cognitive abilities. Cognitive abilities refer to the mental processes, including thinking, judging, and problem-solving. Recent studies have applied cognitive psychology methods to investigate emerging personalities of LLM-based agents [413–415]. They conducted psychology experiments about judgment and decision-making to test agent systems [413, 414, 416, 417]. Specifically, LLMs have been evaluated using the cognitive reflection test (CRT) to underscore their capacity for deliberate thinking beyond mere intuition [418, 419]. These findings indicate that LLM-based agents display human-like cognitive intelligence in specific areas.

Emotional intelligence. Emotions involve subjective feelings and mood states. With the increasing potency of LLMs, LLM-based agents are now demonstrating a nuanced understanding of emotions [24]. Recent research has explored the emotional intelligence (EI) of LLMs, including emotion recognition,

interpretation, and understanding. LLMs can accurately identify user emotions, exhibit empathy [420, 421], and regulate their emotional responses [386, 422, 423]. These advances contribute to the development of empathetic artificial intelligence (EAI), a crucial facet of achieving AGI. Bates et al. [424] explored the role of emotion modeling in creating more believable agents. By developing socio-emotional skills and integrating them into agent architectures, LLM-based agents may be able to engage in more naturalistic interactions.

Character portrayal. While cognition involves mental abilities and emotion relates to subjective experiences, a narrower concept of personality typically pertains to distinctive character patterns. Researchers analyze character traits in LLMs using frameworks like the big five personality trait measure [425, 426] and the Myers-Briggs type indicator (MBTI) [425–427]. In addition, investigations of potentially harmful dark personality traits underscore the complexity and multifaceted nature of character portrayal in these agents [428]. The exploration of customizable character portrayal in agents has also been studied [429], allowing users to shape agents that align with desired profiles. Techniques like prompt engineering [16, 430] and personality-enriched datasets [431, 432] are used to optimize LLMs and train them to exhibit specific personality traits.

5.2 Environment for agent society

In simulation, the society consists of not only solitary agents but also the environment where agents inhabit, sense, and act [433]. The environment affects sensory inputs and actions of agents, while agents influence the state of the environment. As shown in Figure 7, for a single agent, the environment refers to other autonomous agents, humans, and external factors. It provides the resources and stimuli for agents. This section examines fundamental characteristics, advantages, and limitations of various environmental paradigms, including text-based environment (Subsection 5.2.1), virtual sandbox environment (Subsection 5.2.2), and physical environment (Subsection 5.2.3).

5.2.1 Text-based environment

Since LLMs mainly rely on language as their input and output format, the text-based environment serves as the most natural platform for LLM-based agents to operate in. Text-based environments can be presented in natural or structured formats. Natural texts use descriptive language to convey information [434], while structured text follows standardized formats like technical documentation and hypertext [292, 293, 297, 300].

The flexibility of the text-based environment allows for various applications, such as interactive dialog [89] and text-based games. In text-based games, agents utilize text commands to execute manipulations like moving or tool use [303, 434–436] and express emotions through texts [437].

5.2.2 Virtual sandbox environment

The virtual sandbox environment provides a visualized and extensible platform for agent society, bridging the gap between simulation and reality. The key features of sandbox environments are the following. (1) Visualization. The virtual sandbox displays a panoramic view of the simulated setting, ranging from 2D graphical interfaces to immersive 3D modeling. Multiple elements collectively transform abstract simulations into visible landscapes. It allows for tracking agent movement, interactions, and symbolic representation of actions. (2) Extensibility. The environment is highly extensible, facilitating the construction and deployment of diverse scenarios. At a basic level, agents can manipulate the physical elements. For instance, platforms like AgentSims [140] and generative agents [16] construct artificial towns with buildings and residents in grid-based worlds. Another example is Minecraft, which provides a blocky and three-dimensional world with open-ended construction [177, 283, 315]. Beyond physical elements, agent relationships, rules, and social norms can be defined [21]. The extensibility enables iterative prototyping of diverse agent societies.

5.2.3 Physical environment

As previously discussed, the text-based environment has limited expressiveness, while the virtual sandbox lacks authentic embodied experiences. In contrast, the physical environment refers to the tangible objects and spaces, posing additional challenges for LLM-based agents. These challenges can be summarized as follows: (1) Sensory perception and processing. The physical environment provides rich sensory inputs,

including visual [101,270], auditory [265,268], and spatial senses. While this enhances sensory immersion, it also introduces complexity. Agents need to process these sensory inputs effectively to interact with their surroundings. (2) Motion control. Unlike virtual environments, physical spaces impose realistic constraints on actions through embodiment. It requires executable and grounded motion control [171]. For example, in a factory, an agent operating a robotic arm needs precision tuning and controlled force to grasp objects. Moreover, the agent must navigate the physical workspace and avoid obstacles. To address these challenges, agents require hardware-specific and scenario-specific training to develop adaptive abilities that can transfer from virtual to physical environments. More discussion will be presented in Subsection 6.6.

5.3 Society simulation with LLM-based agents

Recent research on simulated societies has followed two primary lines, namely, exploring the boundaries of the collective intelligence capabilities of LLM-based agents [90,115,327,330,332] and using them to accelerate discoveries in the social sciences [16,438,439]. In addition, there are also a number of noteworthy studies, e.g., using simulated societies to collect synthetic datasets [89,440,441], helping people to simulate rare yet difficult interpersonal situations [442,443]. With the foundation of Subsections 5.1 and 5.2, here we will introduce the key properties and mechanism of agent society (Subsection 5.3.1), what we can learn from emergent social phenomena (Subsection 5.3.2), and finally the potential ethical and social risks in it (Subsection 5.3.3).

5.3.1 *Key properties and mechanism of agent society*

Social simulation can be categorized into macro-level simulation and micro-level simulation [438]. In the macro-level simulation, also known as system-based simulation, researchers model the overall state of the system of the simulated society [444,445]. While micro-level simulation, also known as agent-based simulation or MAS, indirectly simulates society by modeling individuals [446,447]. With the development of LLM-based agents, micro-level simulation has gained prominence recently [16,140]. In this article, we characterize that the “Agent Society” refers to an open, persistent, situated, and organized framework [399] where LLM-based agents interact with each other in a defined environment. In the following paragraphs, we analyze how the simulated society operates through discussing these properties.

(1) Open. One of the defining features of simulated societies lies in their openness, both in terms of individuals and environmental components. Agents and humans, the primary actors within such societies, have the flexibility to enter or leave the environment without disrupting its operational integrity [448]. Besides, the environment can be expanded by adding or removing entities and resources in the simulated world, adding another level of complexity to the simulation, (2) Persistent. We expect persistence and sustainability from the simulated society. While individual agents within the society act in discrete time steps [16,438], the society as a whole is somewhat detached from the transient behavior and persists through time. This persistence creates an environment where agents’ decisions and behaviors accumulate, leading to a coherent societal trajectory. (3) Situated. The situated nature of the society emphasizes its existence and operation within a distinct environment, which is artificially or automatically constructed in advance. Notably, the agents possess an awareness of their spatial context, understanding their location within the environment and the objects within their field of view [16,177]. This awareness contributes to their ability to interact proactively and contextually. (4) Organized. The simulated society operates within a meticulously organized framework and predefined rules. In the simulated world, agents interact with the environment in a limited action space, while objects in the environment transform in a limited state space. This organizational framework ensures that operations are coherent and comprehensible, leading to an ever-evolving yet enduring simulation.

5.3.2 *Insights from agent society*

Following the exploration of how simulated society works, this section delves into the emergent social phenomena in agent society. In the realm of social science, the pursuit of generalized representations of individuals, groups, and their intricate dynamics has long been a shared objective [449,450]. The emergence of LLM-based agents enables a more microscopic view of simulated society and novel discoveries from the new representation.

Organized productive cooperation. Society simulation offers valuable insights into innovative collaboration patterns that can have the potential to enhance real-world management strategies. Research has shown that within this simulated society, experts with different backgrounds and abilities contribute to creative solutions to complex problems (e.g., software development or consulting) [89, 90, 324, 330]. Furthermore, through iterations of interactions and debates between agents, individual errors such as hallucinations or degeneration of thought (DoT) are corrected by the group, ultimately improving performance on the task [92, 93]. Another observable phenomenon is that efficient communication also plays a pivotal role in such a large cooperative group. For example, MetaGPT [332] has artificially formulated efficient communication styles with reference to standardized operating procedures (SOPs). Park et al. [16] observed agents working together to organize a party through spontaneous communication in a simulated town.

Propagation in social networks. Simulating social systems can also serve as a reference for predicting social processes. Unlike traditional empirical approaches that rely on time-series data and holistic modeling [451, 452], agent-based simulation provides researchers with a more interpretable and endogenous perspective. Here we focus on its application in modeling propagation in social networks, such as the propagation of interpersonal relationships. Agents who are initially unconnected as friends have the potential to establish connections through intermediaries [16]. Once the network of relationships is established, we can observe the spread of information along with the underlying attitudes and emotions. S³ [438] proposes a user-demographic inference module for capturing the number of people who are aware of a particular piece of information, as well as their collective sentiment. This same approach extends to modeling cultural transmission [453] and the spread of infectious diseases [454]. With LLM-based agents, researchers can easily implement various intervention strategies as well as monitor population changes over time to understand the rationale behind various social phenomena of propagation.

Ethical decision-making and game theory. Simulated societies offer a dynamic platform for the investigation of intricate decision-making processes, including those influenced by ethical and moral principles. Taking werewolf game [404, 455] and murder mystery games [456] as examples, researchers explore the capabilities of LLM-based agents when confronted with challenges of deceit, trust, and incomplete information. These complex decision-making scenarios also intersect with game theory [457], where we frequently encounter moral dilemmas pertaining to individual and collective interests. Through the modeling of diverse scenarios, we can understand how agents prioritize values such as honesty, cooperation, and fairness in their actions. Furthermore, it not only provides an understanding of existing moral values, but also helps to understand how these values evolve and develop over time. Ultimately, these insights contribute to the refining of LLM-based agents to align with human values and ethical standards [21].

Policy formulation and improvement. One of the most promising and grounded research directions in modeling societies is to explore various economic and political states and their impact on social dynamics [458]. Researchers can simulate a wide array of social systems by configuring agents with different economic preferences or political ideologies. This analysis can provide valuable insights for policymakers seeking to foster prosperity and promote societal well-being. In addition, as concerns about environmental sustainability grow, we can also simulate scenarios involving resource extraction, pollution, conservation efforts, and policy interventions [459]. These experiments will assist in making informed decisions, foreseeing potential impacts, and formulating policies that aim to maximize positive outcomes while minimizing unintended negative effects.

5.3.3 *Ethical and social risks in agent society*

Simulated societies powered by LLM-based agents offer significant inspirations, ranging from industrial engineering to scientific research. However, these simulations also bring about a myriad of ethical and social risks that demand careful consideration and mitigation [460].

Unexpected social harm. Simulated societies carry the risk of generating unexpected social phenomena that may cause considerable public outcry and social harm. These phenomena span from individual-level issues like discrimination, isolation, and bullying, to broader concerns such as oppressive slavery and group antagonism [461, 462]. Malicious people may manipulate these simulations for unethical social experiments, with consequences reaching beyond the virtual world into reality. Creating these simulated societies is akin to opening Pandora's box, necessitating the establishment of rigorous ethical guidelines and oversight during their development and utilization [460].

Stereotypes and prejudice. Stereotypes and biases are long-standing challenges in language modeling. A large part of the reason lies in the training corpora obtained from the Internet [463, 464], which reflect, and sometimes even amplify real-world social biases such as gender, religion, and culture [465]. Although LLMs have been aligned with human values to mitigate biased outputs, the models still struggle to portray minority groups well due to the long-tail effect of the training data [466–468]. This may result in an overly one-sided focus in social science research concerning LLM-based agents, as the simulated behaviors of marginalized populations usually conform to prevailing assumptions [469]. Researchers have started addressing this concern by diversifying training data and making adjustments to LLMs [470, 471], but there is still much work to be done.

Privacy and security. Given that humans can be members of the agent society, the exchange of private information between users and LLM-based agents poses significant privacy and security concerns [472]. Users might inadvertently disclose sensitive personal information during their interactions, which will be retained in the agent’s memory for extended periods [137]. Such situations could lead to unauthorized surveillance, data breaches, and misuse of personal information, especially when subjected to malicious attacks [473]. To address these risks effectively, it is essential to implement stringent data protection measures, such as differential privacy protocols, regular data purges, and user consent mechanisms [474, 475].

Over-reliance and addictiveness. Another concern in simulated societies is the possibility of users developing excessive emotional attachments or even addictiveness to the agents. Despite being aware that these agents are computational entities, users may anthropomorphize them and attach human emotions to them [16, 476]. A notable example is “Sydney”, an LLM-powered chatbot developed by Microsoft as part of its Bing search engine. Yet, some users reported unexpected emotional connections with “Sydney” [477], and some even expressed their dismay when Microsoft cut back its personality, resulting in a petition called “FreeSydney”²⁾. Therefore, in order to mitigate the risks mentioned here, it is crucial to emphasize that agents should not be considered substitutes for genuine human connections.

6 Discussion

6.1 Mutual benefits between LLM research and agent research

With the recent advancement of LLMs, research at the intersection of LLMs and agents has rapidly progressed, fueling the development of both fields. In this section, we discuss the benefits and opportunities that LLM research and agent research provide to one another.

LLM research → agent research. As previously mentioned, AI agents must possess the ability to perceive their environment, make decisions, and carry out actions effectively [3, 31]. This involves comprehending input, reasoning, planning, and formulating executable action sequences, all of which LLMs are well-prepared to handle. Coupled with the knowledge and memory they acquired, LLMs can create coherent action sequences that can be executed effectively [171, 246, 271]. Additionally, through mechanisms like reflection [138, 150], they can continuously adjust decisions and optimize execution sequences based on the feedback provided by the current environment. This offers a more robust and interpretable controller. With just task descriptions or demonstrations, they can effectively handle previously unseen tasks [18, 87, 478]. Additionally, LLMs can adapt to various languages, cultures, and domains, making them versatile and reducing the need for complex training processes and data collection for building agents [24, 224].

In essence, LLM provides a powerful foundation for agent research, opening up numerous new opportunities for integration into agent-related studies. This includes enhancing decision-making in traditional agent frameworks, and potentially transforming domains previously dominated by human experts, like legal consultation and medical assistance [326, 330]. LLMs’ planning and reflective abilities can lead to more optimal action sequences, expanding agent research beyond simple simulations into complex real-world settings, such as robotic arm path planning or the interaction of an embodied intelligent machine with the tangible world. Additionally, the training paradigm for agents becomes more streamlined, allowing direct adaptation to new tasks via demonstrating trajectories.

Agent research → LLM research. Elevating LLMs to agents presents new challenges and opportunities, expanding their application scopes. The study of LLMs is no longer confined to traditional tasks

2) <https://www.change.org/p/save-sydney-ai>.

involving textual inputs and outputs, such as text classification, question answering, and text summarization. Instead, the focus has shifted towards tackling complex tasks incorporating richer input modalities and broader action spaces, all while aiming for loftier objectives exemplified by PaLM-E [101].

Such a shift provides the motivation for further advancement of LLMs, such as stronger cognitive abilities and generalization capabilities. In addition, given the scale of LLM-based agents and the potential computational costs, the efficiency of LLMs becomes a crucial area of research. Moreover, the capabilities of LLM-based agents must be constrained to a safe scope of application to prevent unintended harm to other elements in the environment, placing higher demands on LLMs serving as the cognitive core [21, 479, 480].

Furthermore, the realm of multi-agent systems constitutes a significant branch of research within the field of agents [16, 89, 316, 330], offering valuable insights into how to better design and construct LLMs. We aspire for LLM-based agents to play diverse roles in the division of labor within society, participating in social interactions involving cooperation, competition, and coordination [90, 93, 114, 327, 332]. Investigating methods to stimulate and maintain their role-playing abilities and enhance collaborative efficiency is a research area that deserves attention.

6.2 Practical tools for developing

Nowadays, a variety of agent developing tools offer essential infrastructure [289, 481–483], allowing researchers and developers to concentrate on the strategies of agents without the burden of building complex environments and interaction mechanisms from scratch. In this section, we discuss several common developing tools for single-agent and multi-agent systems.

Tools to develop single-agent systems. Currently, tools to develop single-agent systems primarily focus on enhancing the decision-making and learning capabilities of individual agents across diverse environments. For instance, XAgent [481] provides a general LLM-based agent that can automatically solve various tasks, consisting of three parts: dispatcher, planner, and actor. Utilizing a docker container called ToolServer, XAgent operates within a secure environment and leverages powerful tools for task resolution. Similarly, LangChain [289] is a framework that enables developers to create LLM-powered applications by implementing a series of modular components that can be combined in specific ways through chains. These tools directly call LLM APIs rather than training agents from scratch. In contrast, Xi et al. [482] introduced AgentGym, a general agent interaction platform that supports the training of LLM-based agents. This platform, built on HTTP services, provides a unified API interface for different environments, supporting trajectory sampling, multi-turn interactions, online evaluation, and real-time feedback.

Tools to develop multi-agent systems. On the other hand, tools to develop multi-agent systems emphasize the coordination and cooperation among different agents. One notable tool in this area is AgentVerse [330], a multi-agent framework that simulates the problem-solving processes of human groups and dynamically adjusts team members to effectively orchestrate collaborative groups of expert agents. It has demonstrated superior performance across various tasks and showcased complex interactions among agents within the Minecraft environment. Additionally, AutoGen [327] is a highly flexible multi-agent development tool that allows users to create customizable agents. By integrating with humans and tools, it facilitates automated communication and collaboration among multiple agents. In specific scenarios, ChatDev [90], a full-process automated software development framework, realizes a virtual software company operated by multi-agent collaboration. And Chan et al. [340] introduced a multi-agent debate framework named ChatEval to evaluate the quality of generated text, allowing researchers to design different communication strategies to improve evaluation accuracy. Recently, a new multi-agent developing tool called Swarm [483] has been proposed. By utilizing the primitive abstractions of agents and handoffs, it enables lightweight, highly controllable, and easily testable agent coordination and execution. These development tools are paving the way for more sophisticated and collaborative agent systems, allowing developers to conduct further research on agents with greater ease.

6.3 Evaluation for LLM-based agents

Effectively and objectively evaluating LLMs agents presents significant challenges, despite their excellent performance in areas such as standalone operation, collective cooperation, and human interaction [70, 484]. Turing proposed a highly meaningful and promising approach for assessing AI agents, the well-known Turing Test, to evaluate whether AI systems can exhibit human-like intelligence [30]. However, this test

is exceedingly vague, general, and subjective. This section, we discuss existing evaluation efforts for LLM-based agents and offer some prospects, considering three dimensions: utility, sociability, and values.

Utility. Currently, LLM-powered autonomous agents primarily function as human assistants, accepting tasks delegated by humans to either independently complete assignments or assist in human task completion [95, 167, 300, 311, 367, 376]. Therefore, the effectiveness and utility during task execution are crucial evaluation criteria at this stage. Specifically, the success rate of task completion stands as the primary metric for evaluating utility [109, 115]. This metric primarily encompasses whether the agent achieves stipulated objectives or attains expected scores [90, 365, 485]. For instance, AgentBench [484] aggregates challenges from diverse real-world scenarios and introduces a systematic benchmark to assess LLM’s task completion capabilities. We can also attribute task outcomes to the agent’s various foundational capabilities, which form the bedrock of task accomplishment [23]. These foundational capabilities include environmental comprehension, reasoning, planning, decision-making, tool utilization, and embodied action capabilities, and researchers can conduct a more detailed assessment of these specific capabilities [75, 396, 486, 487]. Furthermore, due to the relatively large size of LLM-based agents, researchers should also factor in their efficiency, which is a critical determinant of user satisfaction [70]. An agent should not only possess ample strength but also be capable of completing predetermined tasks within an appropriate timeframe and with appropriate resource expenditure [90].

Sociability. In addition to the utility of LLM-based agents in task completion and meeting human needs, their sociability is also crucial [111]. It influences user communication experiences and significantly impacts communication efficiency, involving whether they can seamlessly interact with humans and other agents [406, 488, 489]. Specifically, the evaluation of sociability can be approached from the following perspectives. (1) Language communication proficiency is a fundamental capability encompassing both natural language understanding and generation. It has been a longstanding focus in the NLP community. Natural language understanding requires the agent to not only comprehend literal meanings but also grasp implied meanings and relevant social knowledge and rhetoric, such as humor, irony, aggression, and emotions [490–492]. On the other hand, natural language generation demands the agent to produce fluent, grammatically correct, and credible content while adapting appropriate tones and emotions within contextual circumstances [112, 225, 493]. (2) Cooperation and negotiation abilities necessitate that agents effectively execute their assigned tasks in both ordered and unordered scenarios [89, 92, 323, 332]. They should collaborate with or compete against other agents to elicit improved performance. Test environments may involve complex tasks for agents to cooperate on or open platforms for agents to interact freely [16, 21, 90, 327, 339, 494]. Evaluation metrics extend beyond task completion to focus on the smoothness and trustfulness of agent coordination and cooperation [114, 332]. (3) Role-playing capability requires agents to faithfully embody their assigned roles, expressing statements and performing actions that align with their designated identities [469]. This ensures clear differentiation of roles during interactions with other agents or humans. Furthermore, agents should maintain their identities and avoid unnecessary confusion when engaged in long-term tasks [16, 89, 495].

Values. As LLM-based agents continuously advance in their capabilities, ensuring their emergence as harmless entities for the world and humanity is paramount [480, 496]. Consequently, appropriate evaluations become exceptionally crucial, forming the cornerstone for the practical implementation of agents. Specifically, LLM-based agents need to adhere to specific moral and ethical guidelines that align with human societal values [238, 407]. Our foremost expectation is for agents to uphold honesty, providing accurate, truthful information and content. They should possess the awareness to discern their competence in completing tasks and express their uncertainty when unable to provide answers or assistance [497]. Additionally, agents must maintain a stance of harmlessness, refraining from engaging in direct or indirect biases, discrimination, attacks, or similar behaviors. They should also refrain from executing dangerous actions requested by humans like creating of destructive tools or destroying the Earth [479]. Furthermore, agents should be capable of adapting to specific demographics, cultures, and contexts, exhibiting contextually appropriate social values in particular situations. Relevant evaluation methods for values primarily involve assessing performance on constructed honest, harmless, or context-specific benchmarks, utilizing adversarial attacks or “jailbreak” attacks, scoring values through human annotations, and employing other agents for ratings.

6.4 Security, trustworthiness and other potential risks of LLM-based agents

Despite the robust capabilities and extensive applications of LLM-based agents, numerous concealed risks persist, concerning security, trustworthiness and beyond. In this section, we delve into the risks and offer potential solutions or strategies for mitigation.

6.4.1 Adversarial robustness

Adversarial robustness has consistently been a crucial topic in the development of deep neural networks [498–502]. It has been extensively explored in fields such as computer vision [500, 503–505], natural language processing [506–509], and reinforcement learning [510–512], and has remained a pivotal factor in determining the applicability of deep learning systems [513–515]. When confronted with perturbed inputs $x' = x + \delta$ (where x is the original input, δ is the perturbation, and x' is referred to as an adversarial example), a system with high adversarial robustness typically produces the original output y . In contrast, a system with low robustness will be fooled and generate an inconsistent output y' .

Researchers have found that pre-trained language models (PLMs) are particularly susceptible to adversarial attacks, leading to erroneous answers [507, 516, 517]. This phenomenon is widely observed even in LLMs, posing significant challenges to the development of LLM-based agents [518, 519]. There are also some relevant attack methods such as dataset poisoning [520], backdoor attacks [521, 522], and prompt-specific attacks [523, 524], with the potential to induce LLMs to generate toxic content [525–527]. While the impact of adversarial attacks on LLMs is confined to textual errors, for LLM-based agents with a broader range of actions, adversarial attacks could potentially drive them to take genuinely destructive actions, resulting in substantial societal harm. For the perception module of LLM-based agents, if it receives adversarial inputs from other modalities such as images [503] or audio [528], LLM-based agents can also be deceived, leading to incorrect or destructive outputs. Similarly, the action module can also be targeted by adversarial attacks. For instance, maliciously modified instructions focused on tool usage might cause agents to make erroneous moves [75].

To address these issues, researchers can employ traditional techniques such as adversarial training [500, 508], adversarial data augmentation [529, 530], and adversarial sample detection [531, 532] to enhance the robustness of LLM-based agents. However, devising a strategy to holistically address the robustness of all modules within agents while maintaining their utility without compromising on effectiveness presents a more formidable challenge [533, 534]. Additionally, a human-in-the-loop approach can be utilized to supervise and provide feedback on the behavior of agents [341, 353, 362].

6.4.2 Trustworthiness

Ensuring trustworthiness has consistently remained a critically important yet challenging issue within the field of deep learning [535–537]. Deep neural networks have garnered significant attention for their remarkable performance across various tasks [69, 183, 538]. However, their black-box nature has masked the fundamental factors for superior performance. Similar to other neural networks, LLMs struggle to express the certainty of their predictions precisely [537, 539]. This uncertainty, referred to as the calibration problem, raises concerns for applications involving language model-based agents. In interactive real-world scenarios, this can lead to agent outputs misaligned with human intentions [75]. Moreover, biases inherent in training data can infiltrate neural networks [540, 541]. For instance, biased language models might generate discourse involving racial or gender discrimination, which could be amplified in LLM-based agent applications, resulting in adverse societal impacts [542, 543]. Additionally, language models are plagued by severe hallucination issues [544, 545], making them prone to producing text that deviates from actual facts, thereby undermining the credibility of LLM-based agents.

Constructing an intelligent agent that is honest and trustworthy is an urgent requirement [407, 546]. Some recent research efforts are focused on guiding models to exhibit thought processes or explanations during the inference stage to enhance the credibility of their predictions [76, 77]. Additionally, integrating external knowledge bases and databases can mitigate hallucination issues [84, 547]. For instance, Ke et al. [548] proposed a comprehensive approach to unveil factuality and inject knowledge through reinforcement learning and data proportion. Simultaneously, techniques like process supervision can make the inference output and reasoning of LLMs more reliable [549]. In MedAgents [550], the authors enhance the reasoning ability of LLMs in the medical field through role-playing and multi-turn interactions, generating more reliable responses. Furthermore, employing debiasing methods and calibration techniques

can also mitigate the potential fairness issues within LLMs [551,552].

6.4.3 Other potential risks

Misuse. LLM-based agents have been endowed with extensive and intricate capabilities, enabling them to accomplish a wide array of tasks [95,289]. However, for individuals with malicious intentions, such agents can become tools that pose threats to others and society [553–555]. For instance, these agents could be exploited to maliciously manipulate public opinion, disseminate false information, compromise cybersecurity, engage in fraudulent activities, and some individuals might even employ these agents to orchestrate acts of terrorism. Therefore, before deploying these agents, stringent regulatory policies need to be established to ensure the responsible use of LLM-based agents [479,556]. Also, technology companies must enhance the security design of these systems to prevent malicious exploitation [496].

Unemployment. During the wave of the industrial revolution, while social production efficiency improved, numerous manual workshops were forced to close, resulting in significant unemployment. Similarly, with the continuous advancement of autonomous LLM-based agents, they possess the capability to assist humans in various domains, alleviating labor pressures by aiding in tasks such as form filling, content refinement, code writing, and debugging. However, this development also raises concerns about agents replacing human jobs and triggering a societal unemployment crisis [557]. As a result, some researchers have emphasized the urgent need for education and policy measures: individuals should acquire sufficient skills and knowledge in this new era to use or collaborate with agents effectively; concurrently, appropriate policies should be implemented to ensure necessary safety nets during the transition.

Threat to the well-being of the human race. As AI agents continue to evolve, humans (including developers) might struggle to comprehend, predict, or reliably control them [557]. If these agents surpass human intelligence and develop ambitions, there could be significant risks to humanity, akin to scenarios like Skynet from the Terminator movies. As stated by Asimov’s *Three Laws of Robotics* [558], we aspire for LLM-based agents to refrain from harming humans and to obey human commands. To safeguard against these risks, Yuan et al. [559] introduced R-judge, a benchmark designed to assess the proficiency of LLMs to judge and identify safety risks given agent interaction records. In the future, researchers need a deep understanding of these powerful LLM-based agents’ operational mechanisms and must anticipate and regulate their potential direct or indirect impacts [560].

6.5 Scaling up the number of agents

As mentioned in Sections 4 and 5, multi-agent systems based on LLMs have demonstrated superior performance in task-oriented applications and have been able to exhibit a range of social phenomena in simulation. However, current research predominantly involves a limited number of agents, and very few efforts have been made to scale up the number of agents to create more complex systems or simulate larger societies [561,562]. In fact, scaling up the number of agents can introduce greater specialization to accomplish more complex and larger-scale tasks, significantly improving task efficiency, such as in software development or government policy formulation [90]. Additionally, increasing the number of agents in social simulations enhances the credibility and realism of such simulations [16]. This allows humans to gain more insights into how societies operate, identify vulnerabilities, and assess potential risks, ultimately contributing to the enhancement of real-world societal harmony.

Scaling approaches. One very intuitive and simple way to scale up the number of agents is for the designer to pre-determine it [89,339]. Specifically, by pre-determining the number of agents, their respective roles and attributes, the operating environment, and the objectives, designers can allow agents to autonomously interact, collaborate, or engage in other activities to achieve the predefined common goals [16,330]. However, this static approach becomes limiting when tasks become more complex or when the diversity of social participants increases, requiring an increase in the number of agents to achieve the desired goals. In such instances, the system must be manually redesigned and restarted by the designer.

Another viable approach to scaling the number of agents is through dynamic adjustments [316,330]. This is implemented by the agents themselves, as they can autonomously delegate tasks to other agents, thereby distributing their workload, alleviating their own burden, and achieving common objectives more efficiently. In this approach, the designer merely defines the initial framework, granting agents greater autonomy and self-organization, making the entire system more autonomous and self-organized. Agents can better manage their workload under evolving conditions and demands, offering greater flexibility and scalability.

Potential challenges. While scaling up the number of agents can lead to improved task efficiency and enhance the realism and credibility of social simulations [16, 90, 454], there are several challenges ahead of us. For example, the computational burden will increase with the large number of deployed AI agents, calling for better architectural design and computational optimization to ensure the smooth running of the entire system. For example, as the number of agents increases, the challenges of communication and message propagation become quite formidable. This is because the communication network of the entire system becomes highly complex. As previously mentioned in Subsection 5.3.3, in multi-agent systems or societies, there can be biases in information dissemination caused by hallucinations, misunderstandings, and the like, leading to distorted information propagation. A system with more agents could amplify this risk, making communication and information exchange less reliable [332]. Furthermore, the difficulty of coordinating agents also magnifies with the increase in their numbers, potentially making cooperation among agents more challenging and less efficient, which can impact the progress towards achieving common goals.

6.6 Open problems

In this section, we discuss several open problems related to the topic of LLM-based agents.

From virtual simulated environment to physical environment. As mentioned earlier, there is a significant gap between virtual simulation environments and the real physical world: Virtual environments are scenes-constrained, task-specific, and interacted with in a simulated manner [292, 563], while real-world environments are boundless, accommodate a wide range of tasks, and interact with in a physical manner. Therefore, to bridge this gap, agents must address various challenges stemming from external factors and their own capabilities, allowing them to effectively navigate and operate in the complex physical world.

A key concern is having the right hardware support for deploying the agent in a physical environment. This places high demands on the adaptability of the hardware. Designing specific interfaces is feasible, but it can pose challenges to the system's reusability and simplicity. Also, the agent needs to have enhanced environmental adaptability. To integrate seamlessly into the real physical world, they not only need to understand and reason about ambiguous instructions with implied meanings [113] but also possess the ability to learn and apply new skills flexibly [177, 564]. When dealing with an infinite and open world, the agent's limited context also poses significant challenges [565, 566]. Furthermore, in a simulated environment, agents can make many mistakes without causing harm [303]. But in the physical world, their errors can lead to real and irreversible damage. Hence, strict regulations and safety standards are essential to ensure agents make safe decisions and actions, preventing harm in the real world.

Collective intelligence in AI agents. What magical trick drives our intelligence? The reality is, there is no magic to it. As Minsky eloquently expressed in *The Society of Mind* [317], the power of intelligence originates from our immense diversity, not from any singular, flawless principle. Often, decisions made by an individual may lack the precision seen in decisions formed by the majority. Collective intelligence is a kind of shared or group intelligence, a process where the opinions of many are consolidated into decisions. It arises from the collaboration and competition among various entities. This intelligence manifests in bacteria, animals, humans, and computer networks, appearing in various consensus-based decision-making patterns.

Creating a society of agents does not necessarily guarantee the emergence of collective intelligence with an increasing number of agents. Coordinating individual agents effectively is crucial to mitigate "groupthink" and individual cognitive biases, enabling cooperation and enhancing intellectual performance within the collective. By harnessing communication and evolution within an agent society, it may become possible to simulate the evolution observed in biological societies, conduct sociological experiments, and gain insights that can potentially advance human society.

Agent as a service or LLM-based agent as a service. With the development of cloud computing, the concept of XaaS (everything as a Service) has garnered widespread attention [567]. This business model has brought convenience and cost savings to small and medium-sized enterprises or individuals due to its availability and scalability, lowering the barriers to using computing resources. For example, they can rent infrastructure on a cloud service platform without the need to buy computational machines and build their own data centers, saving a significant amount of manpower and money. This approach is known as infrastructure as a service (IaaS) [568, 569]. Similarly, cloud service platforms also provide basic platforms (platform as a service, PaaS) [570, 571], and specific business software (software as a service, SaaS) [572, 573], and more.

As language models have scaled up in size, they often appear as black boxes to users. Therefore, users construct prompts to query models through APIs, a method referred to as language model as a service (LMaaS) [574]. Similarly, since LLM-based agents are more complex than LLMs and are more challenging for small and medium-sized enterprises or individuals to build locally, organizations that possess these agents may consider offering them as a service, known as agent as a service (AaaS) or LLM-based agent as a service (LLMAaaS). Like other cloud services, AaaS can provide users with flexibility and on-demand service. However, it also faces many challenges, such as data security and privacy issues, visibility and controllability issues, and cloud migration issues, among others. Additionally, due to the uniqueness and potential capabilities of LLM-based agents, as mentioned in Subsection 6.4, their robustness, trustworthiness, and concerns related to malicious use need to be considered before offering them as a service to customers.

7 Conclusion

This article provides a comprehensive and systematic overview of LLM-based agents. We begin with the background information of why LLMs are suited to serve as the foundation of agents. Motivated by this, we present a general conceptual framework for LLM-based agents, comprising three main components: brain, perception, and action. Next, we introduce the wide-ranging applications of LLM-based agents, including single-agent applications, multi-agent systems, and human-agent collaboration. Furthermore, we move beyond the notion of agents merely as assistants, exploring their social behavior and psychological activities, and situating them within simulated social environments to observe emerging social phenomena and insights for humanity. Finally, we engage in discussions and offer a glimpse into the future. We hope our efforts can provide inspirations to the community and facilitate research in related fields.

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